



Electronic Invoice Management System (EIMS)

Ministry of Revenues
(MoR)

COMPLIANCE DOCUMENT

Version: 1.0

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A. ABBREVIATIONS/ACRONYMS

Table 1 : Abbreviations/Acronyms

Abbreviation	Explanation
MoR	Ministry of Revenue
EIMS	Electronic Invoice Management System
TP	Taxpayer
TA	Tax Authority
PKI	Public Key infrastructure

1. SECTION ONE

1.1 INTRODUCTION

The Minister of Revenue is actively working on implementing an advanced electronic invoice management system designed to modernize the invoicing process, enhance transparency, and uphold integrity in financial transactions. This initiative aims to significantly improve operational efficiency in tax collection while ensuring a more seamless and reliable approach to managing invoices. The introduction of this system is expected to bring substantial benefits to both taxpayers and the Ministry.

For taxpayers, the system simplifies adherence to tax regulations, minimizes paperwork, and reduces administrative burdens associated with financial transactions. By automating and streamlining processes, it eliminates many of the complexities traditionally involved in invoicing and tax compliance. As a result, businesses and individuals can manage their tax-related responsibilities with greater ease, accuracy, and efficiency.

From the Ministry's perspective, the system plays a crucial role in strengthening tax compliance by ensuring that financial transactions are more transparent and traceable. Additionally, it facilitates more effective and timely tax collection, reducing the risk of tax evasion and improving overall revenue generation. By leveraging technology, the Ministry is taking a proactive step toward a more efficient, transparent, and accountable tax administration system that benefits all stakeholders involved.

1.2 STRUCTURE OF THE DOCUMENT

The document is systematically structured into six well-defined sections, each addressing a key aspect of the Electronic Invoice Management System (EIMS) to ensure clarity and comprehensive understanding.

The first section functions as the introduction, offering a comprehensive overview of the document's purpose, the legal framework guiding the implementation of the Electronic Invoice Management System (EIMS), a technical introduction, and an in-depth explanation of its core processes. This section lays the groundwork for understanding the system, emphasizing its importance in optimizing invoice management and ensuring tax compliance.

The second section presents an in-depth discussion on invoice registration guidelines. It outlines the essential steps, requirements, and procedural details necessary for taxpayers to correctly register invoices within the system. This section ensures that users adhere to compliance standards while minimizing errors in invoice submission.

The third section focuses on the required invoice schema, detailing the specific structure, format, and key data elements that must be included in an invoice. This section ensures that all invoices follow a standardized format, making them compatible with the EIMS and regulatory requirements.

In the fourth section, the document provides a thorough explanation of bulk invoice processing guidelines. It covers the step-by-step procedures for handling multiple invoices efficiently, including technical considerations and best practices to ensure smooth processing for businesses that generate high volumes of invoices.

The fifth section is dedicated to the guidelines for invoice cancellation. It explains the circumstances under which an invoice can be canceled, the procedural requirements for doing so, and the compliance measures that must be followed.

The sixth section provides a detailed explanation of the guidelines for registering the receipt of an invoice that has already been previously registered. It outlines the key steps, necessary documentation, and procedures required for the successful submission of the receipt to the electronic invoice management system.

Finally, the seventh section outlines the security guidelines necessary for the effective operation of both the taxpayers and the EIMS. It highlights the security measures, data protection protocols, and best practices that ensure the system runs smoothly while safeguarding sensitive financial information. This section is crucial in maintaining the integrity, confidentiality, and reliability of the electronic invoicing system.

1.3 PURPOSE OF THE GUIDELINE

To ensure the Electronic Invoice Management System works seamlessly with taxpayers' systems, proper integration based on predefined standards and guidelines is essential. This document is designed to help taxpayers meet the mandatory requirements set by the Ministry of Revenue of the

Federal Government of Ethiopia. It provides detailed guidance on the technical specifications needed for successful integration, ensuring smooth and efficient operation of the system.

1.4 LEGAL FRAMEWORK

The introduction of electronics invoice management system can be outlined to the law which was issued by ----- which enforces an obligation to taxpayers to prepare, send and keep invoices in electronic form across the board. Among other conditions, the main purposes pursued by adopting electronic invoicing are: improvement of compliance by the taxpayers with correct tax reporting; minimal possibilities of fraud owing to improved transparency; and simplification for businesses and government's tax officials. The switch to e-invoicing is part of a wider strategy aimed at automating the collection of taxes and enhancing accountability measures in the administration of taxes to facilitate efficiency and security of financial transactions.

1.5 INTENDED AUDIENCE

This document is intended for software developers, system architects, and QA engineers who are tasked with implementing or testing the integration of systems with the electronics invoice management system. It provides valuable guidance to help these professionals ensure that their systems are correctly integrated and meet the required e-invoicing standards. Whether you're developing the integration or checking its functionality, this document offers the information you need to support the process effectively.

1.6 TECHNICAL INTRODUCTION TO EIMS

This document outlines the technical requirements for the electronic invoice management system, providing key guidelines and specifications for its successful implementation and integration. These comprehensive requirements ensure the efficient operation of all system components. By adhering to them, businesses and taxpayers can seamlessly integrate with the system, ensuring accurate invoicing, regulatory compliance, and streamlined data processing.

Additionally, the document details the API requirements for registering invoice documents within the system. It provides in-depth technical guidelines and protocols to facilitate smooth integration with the Electronic Invoice Management System. The specifications define the structure,

parameters, and data formats needed for the precise submission, validation, and storage of invoices. Compliance with these standards enables businesses to optimize their invoicing processes, meet regulatory requirements, and support real-time data exchange.

Moreover, the document includes API specifications for managing both single and bulk invoice cancellations, as well as verifying invoice legality. It outlines the technical requirements, data formats, and communication protocols essential for accurate and efficient processing. These guidelines help taxpayers integrate cancellation functionalities into their systems, process cancellations based on predefined rules, and ensure compliance with legal and regulatory standards. Adhering to these specifications supports regulatory compliance, improves operational efficiency, and safeguards the integrity of the invoicing process.

Furthermore, this document contains security specifications necessary for the secure and seamless operation of the electronic invoice management system in coordination with the taxpayer's system.

Before initiating operations and exchanging information with the Electronic Invoice Management System through its API, taxpayers must:

- Obtain credentials for their systems from EIMS taxpayer Portal
- Obtain digital certificate from a Certificate Authority as most communications with EIMS require digital signature

1.7 EIMS PROCESS FLOWS

1.7.1 List of Invoice Processes

EIMS is the electronic invoice registration, cancellation and verification system of the Ministry of Revenue of The Federal Democratic Republic of Ethiopia. The main processes of EIMS are:

1. Authenticate with EIMS for API communication
2. Enable taxpayer systems send requests to register invoices via API and receive clearance of invoice with unique Invoice Registration Number (IRN) and QR Code
3. Enable taxpayer systems send request to cancel invoices via API and receive confirmation of cancellation

4. Enable taxpayer systems send bulk registration/cancellation requests via API or File uploads and receive results of their requests via Call Back API or File Downloads

The following sub-sections provide high-level overview of the above process flows which are explained in detail in other related documents that are listed in section 3.0.

1.7.2 Authentication

Taxpayer Systems need to authenticate with EIMS Authentication APIs using AOUTH2 for communicating through REST end points. For authentication taxpayer systems need to be configured with API Key, Client ID and Client Secret which will be provided via secure communication during the on-boarding process. In addition, taxpayer System administrators can change these credentials via the EIMS taxpayer portal.

Upon successful authentication, the Electronic Invoice Management System (EIMS) issues a JWT token with a validity of three hours. Additionally, the Taxpayer System receives a refresh token, allowing it to renew the JWT token without requiring re-authentication. The specifics of this process are outlined in the security specifications section of this document.

The diagram below provides an overview of the authentication process.

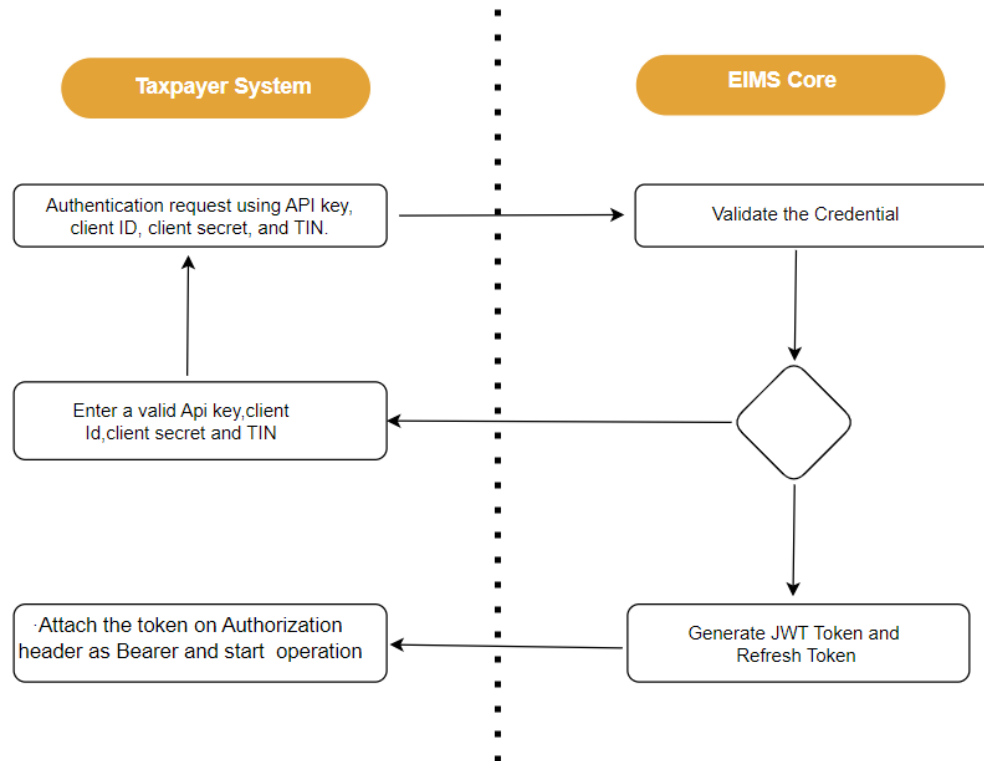


Figure 1 Authentication Process

Table 2 Authentication Services

Service name	Authentication
Access Method	REST API - POST
Current Version	1
Prerequisite	
Expected Results	JWT Token and Refresh Token on successful authentication or wrong credentials.
Data Exchange Request and Response	JSON

Table 3 Token Refresh Services

Service name	Refresh Authentication
Access Method	REST API - POST
Current Version	1
Prerequisite	Authentication of taxpayer system with EIMS
Expected Results	New JWT Token that can be used or failure with not authorized

1.7.3 Invoice Registration Process

Taxpayer systems need to authenticate with EIMS before they can request for invoice document registration. When taxpayer system have obtained required JWT token and refresh token, then it can send a request for invoice registration through the registration API with content that is compliant with the registration API of EIMS. At this stage EIMS will reply with IRN, QR-code and signed invoice if the registration is successful or will replay with error messages

The below diagram shows a high-level representation of invoice registration process.

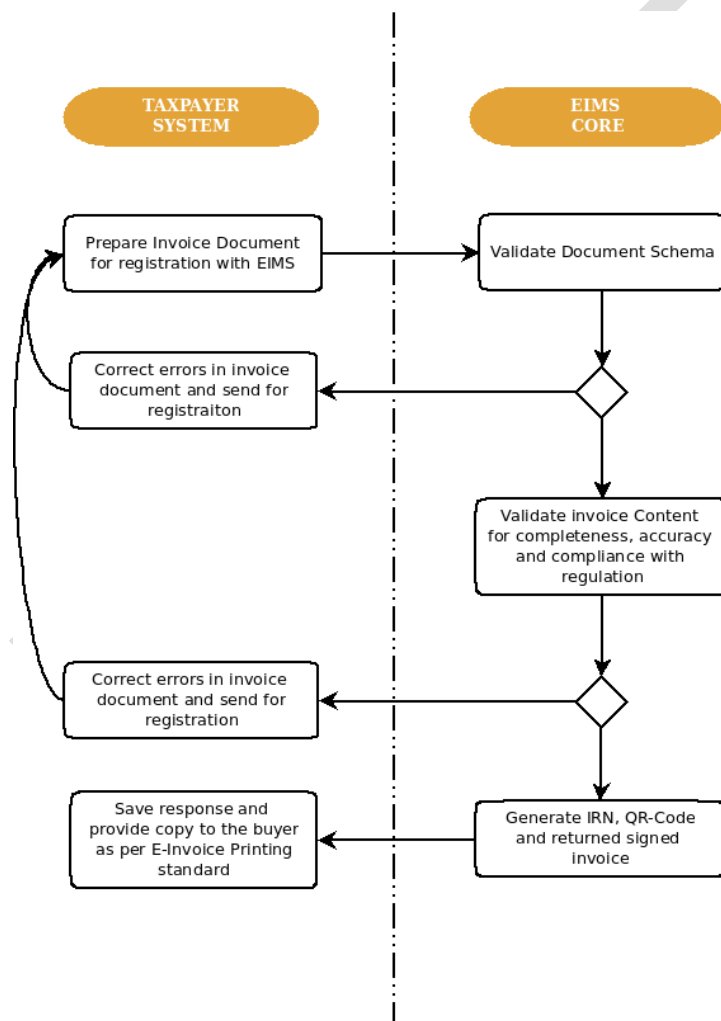


Figure 2 Invoice Registration

Table 4 Invoice Registration Services

Service name	Invoice document registration request
Access Method	REST API
Current Version	1
Prerequisite	Authentication of taxpayer system with EIMS
Expected Results	Success with IRN, QR and Signed Invoice and Failure with one of the standard failure messages
Data Exchange Request and Response	JSON

1.7.4 Invoice Cancellation

Process

Taxpayer systems can submit a request to cancel an invoice, which will be verified by EIMS based on applicable rules and regulations for invoice cancellations. If the request meets the validation criteria, a cancellation confirmation will be issued. Invoice cancellation is only permitted for invoices registered in EIMS and must comply with specific rules, including the maximum allowable time between issuance and cancellation, as well as daily and monthly limits on the number of cancellations.

A cancellation request is automatically processed if it meets the predefined rules and falls within the permitted daily and monthly cancellation limits. However, if the request exceeds these limits, it will be subject to manual review before any further action is taken. This manual review ensures compliance with regulations and prevents excessive or illegal cancellations. The validation process helps maintain the integrity of the system by enforcing restrictions on the number of cancellations allowed within a given period.

Table 5: Cancellation Process

Service name	Invoice cancellation
Access Method	REST API
Current Version	1
Prerequisite	Authentication of taxpayer system with EIMS
Expected Results	Successful cancellation will be notified to the taxpayer system with “Success” response while failure will contain failure reasons
Data Exchange Request and Response	JSON

1.7.5 Batch Registration and Cancellation

EIMS has provision for batch registration and cancellation using two different approaches, though bulk registration/cancellation APIs and through bulk registration/cancellation request files.

1.7.5.1 Bulk Registration and Cancellation API

With bulk registration and cancellation APIs, a number of requests are sent to EIMS and EIMS processes and responds to these requests asynchronously. Due to the nature of asynchronous responses, taxpayer systems will have to make ready a web call back interface that will receive results of the bulk registration/cancellation requests. Different rules of registration as well as cancellation are applied during bulk registration/cancellation than those for single invoice registration/cancellation.

Table 6 : Bulk registration/cancellation service

Service name	Bulk Registration API
Access Method	REST API
Current Version	1
Prerequisite	Authentication of taxpayer system with EIMS
Expected Results	Call back URL of the customer will be invoked with the output of the bulk request
Data Exchange Request and Response	JSON

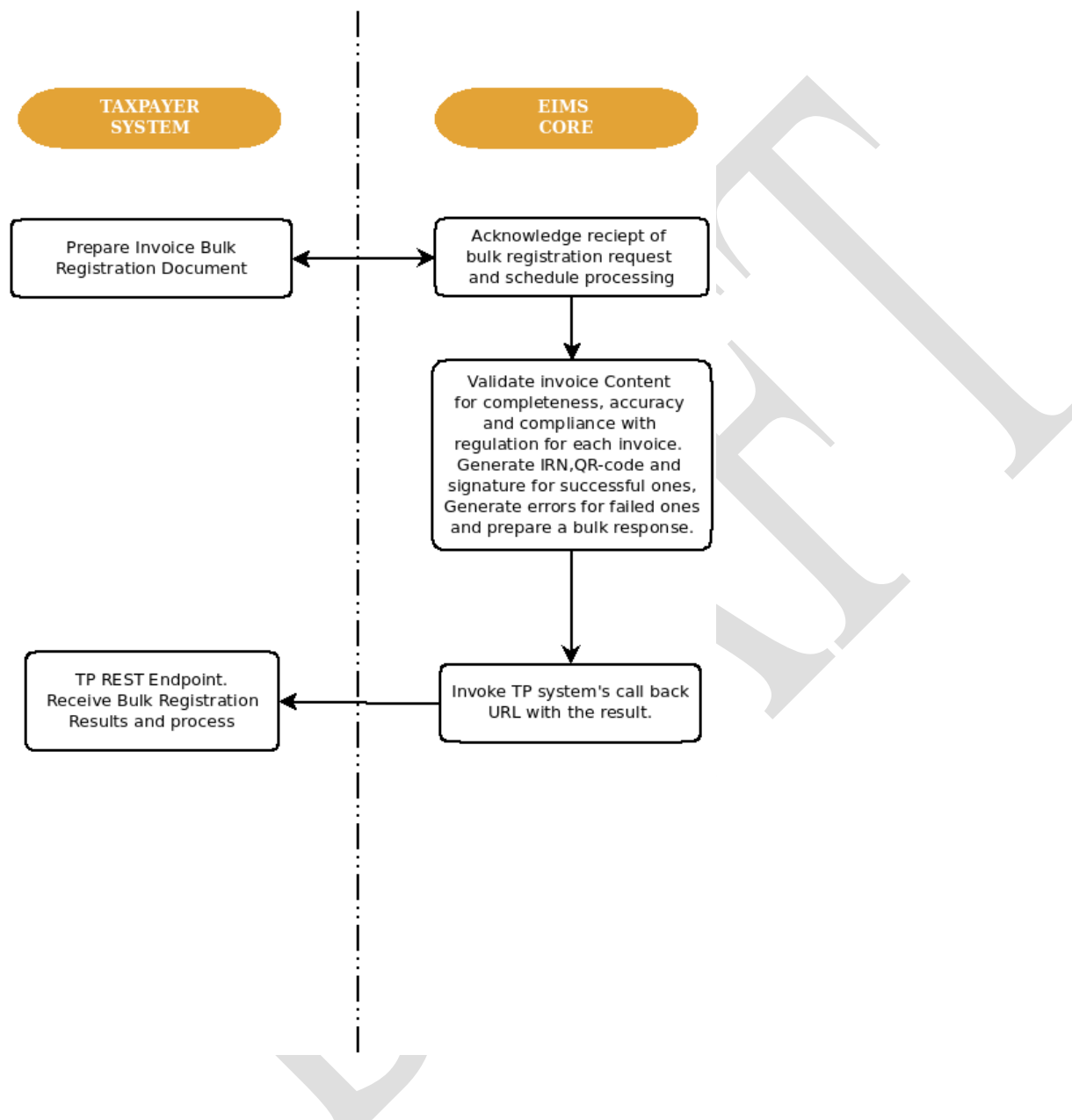


Figure 3: Bulk registration and Cancellation

1.7.5.2 Bulk Registration and Cancellation with File Exchange

For both bulk registration and cancellation, EIMS provides file based request processing. Taxpayers can upload files for registration and cancellation on the EIMS portal which will then be processed

For registration/cancellation and the results can also be download from the portal.

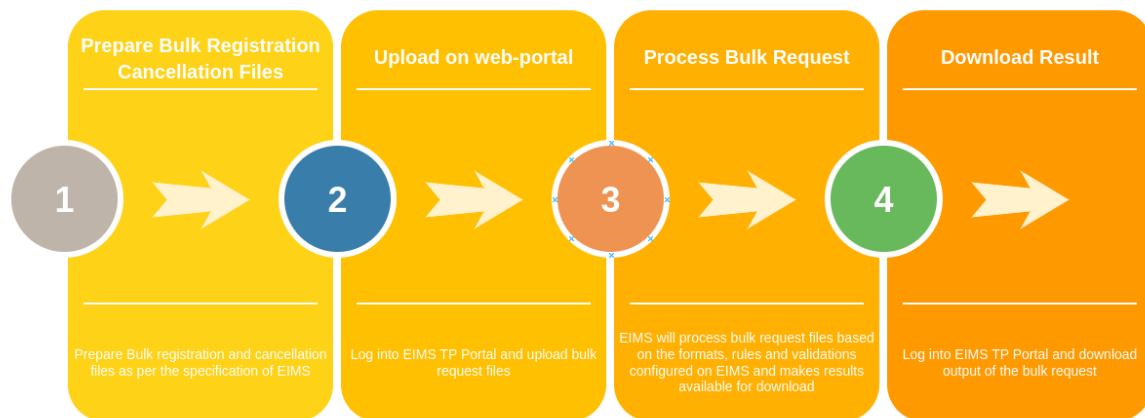


Figure 4: bulk registration and cancelation with file

2 SECTION TWO

2.1 INVOICE REGISTRATION

2.1.1 PURPOSE

A primary function of the Electronic Invoice Management System (EIMS) is invoice registration, which plays a vital role in managing invoices and ensuring tax compliance. This section provides a general overview of how invoice registration requests are submitted to EIMS, emphasizing the overall process rather than the specific attributes of the invoice registration object. For a comprehensive understanding of these attributes, including the necessary fields and their formats, this section offers a detailed guide. This helps taxpayers accurately register invoices in the system while adhering to compliance requirements.

2.1.2 OVERVIEW

This section provides an overview of the integration requirements for submitting a single e-invoice to the Ministry of Revenue (MoR) e-invoice endpoint. It outlines the essential steps and compliance standards needed to ensure a seamless and successful process. This section covers the following key areas:

- **Request JSON Schema:** The structure and validation rules of the e-invoice request payload.
- **Digital Signing Requirements:** How to digitally sign the request and attach the certificate.
- **Response Types:**
 - **Success Response (200):** Returned when an invoice registration is successful.
 - **Schema Validation Error Response (400):** Returned when the request JSON does not meet the schema definition.
 - **Rule Validation Error Response (406):** Returned when the JSON schema is correct but the invoice fails business/rule validation checks.

2.1.3 REQUEST SCHEMA

The request must be structured as a JSON object and include three primary properties: request, signature, and certificate. The request object must follow the specified JSON schema to ensure

proper formatting and data integrity. Additionally, the signature and certificate are essential components that must be included to verify the authenticity of the digital signature. These elements work together to ensure the security and integrity of the request, confirming that it has not been altered and originates from an authorized source.

2.1.3.1 Top Level Structure

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "type": "object",
  "properties": {
    "request": {
      "$id": "http://www.mor.gov.et/einvoice",
      "type": "object",
      ...
      "description": "Invoice number generation schema"
    },
    "signature": {
      "type": "string"
    },
    "certificate": {
      "type": "string"
    }
  },
  "required": ["request", "signature", "certificate"]
}
```

Key Points:

- **Request:** Contains the invoice details including *DocumentDetails*, *BuyerDetails*, *SellerDetails*, *ItemList*, *ValueDetails*, *PaymentDetails*, *SourceSystem*, *ReferenceDetails*, and *TransactionType*. Details of this is provided in subsequent documents.
- **Signature:** A digital signature string generated by the client's private key over the canonicalized request object.
- **Certificate:** A Base64-encoded certificate (e.g., X.509) used to verify the digital signature.

2.1.3.2 Digital Signing Requirements

Canonicalization of Request Object:

- Before signing, the request object must be canonicalized (e.g., using a deterministic JSON serialization without extra whitespaces or formatting differences).

Signature Generation:

- Use the private key associated with the provided certificate. Compute the digital signature (e.g., using RSA or ECDSA) over the canonicalized request JSON (not including the *Signature* and *Certificate* properties).
- Base64-encode the resulting signature.

Attach Signature and Certificate:

- Signature: The Base64-encoded digital signature string.
- Certificate: The Base64-encoded client certificate that corresponds to the private key used for signing.

Validation on the Server Side:

- The server will verify that the signature is valid for the given request using the provided certificate.
- Any mismatch or invalid signature will cause the request to be rejected.

2.1.4 RESPONSES

2.1.4.1 Success Response (200)

When the invoice request is valid, correctly signed, and meets all the necessary business rules, a success response with a status code of 200 will be sent. This response will include the details of the newly registered invoice, including its unique Invoice Reference Number (IRN). The format of this response is shown in the schema below, providing all the necessary information to confirm that the invoice has been successfully registered.

{

"\$schema": "http://json-schema.org/draft-07/schema#",

"type": "object",

"properties": {

 "statusCode": {

 "type": "integer"

 },

 "message": {

 "type": "string"

 },

 "body": {

 "type": "object",

 "properties": {

 "irn": {

 "type": "string"

 },

 "ackDate": {

 "type": "string",

 "format": "date-time"

 },

 "signedQR": {

 "type": "string"

 },

 "status": {

 "type": "string"

 },

 "documentNumber": {

 "type": "string"

 },

 "signedInvoice": {

 "type": "string"

 },

 "invoiceSignature": {

 "type": "string"

 }

 },

```
    "required": ["irn", "ackDate", "signedQR", "status", "documentNumber", "signedInvoice", "invoiceSignature"]
  }
},
"required": ["statusCode", "message", "body"]
}
```

Example Successful Response is show below,

```
{
  "statusCode": 200,
  "message": "SUCCESS",
  "body": {
    "irn": "51fa3144ae45d2a06873a1e81c59ab7412bbd1ba2a52e70f71be347d586a1d9a",
    "ackDate": "2024-12-20T10:18:35.346863361+03:00[Africa/Addis_Ababa]",
    "signedQR":
    "iVBORw0KGgoAAAANSUhEUgAAAMgAAADIAQAAACFI5MzAAAFc0IEQVR4X...",
    "status": "A",
    "documentNumber": "253",
    "signedInvoice":
    "Cax4wttFy8+BWb5z5P+S1mQ0UvJ3SaTDqNSINcASLBNLaoWYtpP3mPVmiePmMYvx...",
  }
}
```

The following table is a description of the content of the response,

Tag	Description	specification
statusCode	Indicates the status of the request, where 200 indicated successful processing in the example above	Numeric value, 200 – successfully processed 400 – schema validation error 406 – rule validation error
message	Associated message	Success, Schema validation error or Rule validation error
body	Object that contains details of the response	Json Object
irn	A hash value of the contents of the invoice document that is registered	String of length 64 characters
signedQR	Base64 representation of QR code which is 600X600, high error correction level (30%) and containing fields as stated in the schema details table.	
signedInvoice	Contents of the registered invoice digest signed by the private key of MoR. This digitally signed string can be verified by tax payers by taking the invoice data in the response (without change) and generating the hash using the public key of MoR.	

2.1.4.2 Schema Validation Error Response (400)

If the request payload does not comply with the JSON Schema (e.g., required fields missing, incorrect data types), a 400 error is returned. This error provides details about which fields failed validation. Below is the JSON schema response when the request for registration encounters schema validation error,

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "type": "object",
  "properties": {
    "statusCode": {
      "type": "integer"
    },
    "message": {
      "type": "string"
    },
    "body": {
      "type": "array",
      "items": {
        "type": "object",
        "properties": {
          "type": {
            "type": "string"
          },
          "code": {
            "type": "string"
          },
          "message": {
            "type": "string"
          },
          "instanceLocation": {
            "type": "string"
          },
          "property": {
            "type": "string"
          },
          "evaluationPath": {
            "type": "string"
          }
        }
      }
    }
  }
}
```



```
    },
    "schemaLocation": {
      "type": "string",
      "format": "uri"
    },
    "messageKey": {
      "type": "string"
    },
    "arguments": {
      "type": "array",
      "items": {
        "type": "string"
      }
    }
  },
  "required": [
    "type",
    "code",
    "message",
    "instanceLocation",
    "property",
    "evaluationPath",
    "schemaLocation",
    "messageKey",
    "arguments"
  ]
}
},
"required": [
  "statusCode",
  "message",
  "body"
]
}
```

Example of Schema Error is shown below

```
{
  "statusCode": 400,
  "message": "SCHEMA ERROR",
  "body": [
    {
      "type": "required",
      "code": "1028",
      "message": "$.DocumentDetails: required property 'DocumentNumber' not found",
      "instanceLocation": "$.DocumentDetails",
      "property": "DocumentNumber",
      "evaluationPath": "$.properties.DocumentDetails.required",
      "schemaLocation": "http://www.mor.gov.et/einvoice#/properties/DocumentDetails/required",
      "messageKey": "required",
      "arguments": [
        "DocumentNumber"
      ]
    }
  ]
}
```

Key Points:

- statusCode: 400 indicates schema validation errors.
- message: "SCHEMA ERROR".
- body: An array of error objects detailing the specific violations:
 - type: The type of validation error.
 - message: Detailed description of the error.
 - property: The property causing the error.
 - schemaLocation: Points to the schema definition that was violated.
 - arguments: Lists the missing or invalid properties.

2.1.4.3 Rule Validation Error Response (406)

If the request passes the schema validation but fails to meet the business rules or logic (such as an invoice date being out of range or invalid reference data), a 406 error response will be returned. This means that while the format of the request is correct, the content doesn't follow the required rules or guidelines. The schema for the invoice rule validation error is shown below, which helps clarify what went wrong and why the request was rejected.

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "type": "object",
  "properties": {
    "statusCode": {
      "type": "integer"
    },
    "message": {
      "type": "string"
    },
    "body": {
      "type": "array",
      "items": {
        "type": "object",
        "properties": {
          "code": {
            "type": "string"
          },
          "property": {
            "type": "string"
          }
        }
      }
    },
    "message": {
      "type": "string"
    }
  }
},
```

```
    "required": ["code", "property", "message"]
  }
}
},
"required": ["statusCode", "message", "body"]
}
```

Example of Rule Validation Error

```
{
  "statusCode": 406,
  "message": "RULE VALIDATION ERROR",
  "body": [
    {
      "code": "###",
      "property": "DocumentDetails.Date",
      "message": "Document creation time should be within ## days"
    },
    {
      "code": "###",
      "property": "DocumentDetails.Date",
      "message": "Document creation time should be within an open tax period."
    },
    {
      "code": "###",
      "property": "ReferenceDetails.PreviousIrn",
      "message": "Previous IRN in ReferenceDetail is not correct."
    },
    {
      "code": "###",
      "property": "valueDetails.ExchangeRate",
      "message": "Exchange rate is required if currency is not ETB."
    }
  ]
}
```

Key Points:

- statusCode: 406 indicates business/rule validation failure.
- message: "RULE VALIDATION ERROR".
- body: An array of rule validation errors, where each error:

2.1.5 INVOICE PRINT/TRANSMISSION FORMATS

- Compact layout for Fiscal print with embedded printers.
- A4 print for systems with regular printers and electronic transmission.

Please refer to the Annex for detail printed invoice format.

2.1.6 LOOKUP VALUE LIST**TransactionType**

S.N	Value	Description	Start Date	End Date
1				
2				
n				

xxxx

S.N	Value	Description	Start Date	End Date
1				
2				
n				

Error List

S.N	Error Code	Status Code	Description	Resolution procedure
1.	4400	401	User not found during authentication	Use correct credentials and try again.
2.	4401	401	Group not found during authentication	Use correct credentials and try again.
3.	4402	401	General authentication attribute not found	Use correct credentials and try again.
4.	4403	401	API key attribute not found during authentication	Use correct credentials and try again.
5.	4404	401	TIN attribute not found during authentication	Use correct credentials and try again.
6.	4405	401	Encryption key attribute not found during authentication	Use correct credentials and try again.
7.	4406	401	Authentication endpoint is not reachable	Please raise ticket with MoR to investigate the issue.
8.	4407	401	Error response from the authentication endpoint	Use correct credentials and try again, in addition raise support ticket with MoR
9.	4408	401	Invalid API key used during login	Use correct credentials and try again.
10.	4409	401	Invalid TIN used during login	Use correct credentials and try again.
11.	4410	401	Invalid password used during login	Use correct credentials and try again.
12.	4411	401	Refresh token endpoint is not reachable	Please raise support ticket with MoR.
13.	4412	401	Error response from the refresh token endpoint	Please raise support ticket with MoR.
14.	4501	401	Error occurred at the authentication entry point	Please raise support ticket with MoR.
15.	4502	401	Access denied error occurred	Re-authenticate and try again. If the issue persists, raise support ticket with MoR.
16.	4503	401	General token-related issue occurred	Re-authenticate and try again. Please raise support ticket with MoR.
17.	4601	400	Encryption issue	Re-authenticate and obtain new encryption key and try again.

18.	4602	400	Decryption issue	Re-authenticate and obtain new encryption key and try again.
19.	4603	400	Issue during TIN extraction	Please raise support ticket with MoR.
20.	4604	400	Issue during username extraction	Please raise support ticket with MoR.
21.	4605	400	Issue during session state extraction	Please raise support ticket with MoR.
22.	4606	400	Unknown error in the encryption/decryption process	Please raise support ticket with MoR.
23.	4607	400	Issue during encryption key extraction	Please raise support ticket with MoR.
24.	67001	400	Unknown certificate validation error	Please raise support ticket with MoR.
25.	67002	400	Error loading the trust store	Please raise support ticket with MoR.
26.	67003	400	Certificate is expired	Please contact the Certificate Authority and obtain fresh certificate before trying again.
27.	67004	400	Certificate's signature is invalid	Please contact the certificate authority to verify certificate validity as well as correctness.
28.	67005	400	OCSP validation service is unreachable	Please raise support ticket with MoR.
29.	67006	400	OCSP URL is not found inside the provided certificate	Please check with the Certificate authority and communicate the error description and raise a support request with the response.
30.	67007	400	Invalid OCSP response	Please raise support ticket with MoR.
31.	67008	400	Certificate has been revoked	Please obtain a fresh certificate from the Certificate Authority and try again.
32.	67009	400	Certificate is not in Base64 encoding format	Provide a correct base64 encoded request
33.	67010	400	Public key used in the certificate is invalid	Provide a correct public certificate based on the private key
34.	67011	400	Error during certificate chain validation	Provide a correct Certificate chain on the request
35.	67012	400	Invalid certificate is provided	Provide a correct Certificate on the request

36.	67013	400	Error during certificate signature validation	Provide a valid signature for the invoice request, recheck if request match with the signature
37.	67014	400	Invalid request object is provided	Provide a correct request body
38.	67015	400	Ownership of the certificate is not valid	Provide a correct Certificate
39.	7030	406	Deviation validator for exchange rate	Provide an acceptable Exchange rate
40.	7029	406	Seller Details Existence Checker	Provide a correct Seller detail
41.	7028	406	Document Type Validator	Provide a correct document type
42.	7027	406	Transaction Type Checker	Provide a correct transaction type
43.	7026	406	Document Date By Type	Provide correctly formatted date
44.	7025	406	Validate Nature of Supply	Provide Correct nature of supply type
45.	7024	406	Tax code validator	Provide Correct tax code
46.	7023	406	Unit of measure	Provide a correct unit of measure
47.	7022	406	Payment mode	Provide payment mode
48.	7021	406	Related Document Validator	Provide a correct related document number from previously registred invoice
49.	7020	406	Credit Debit Validator	Provide a correct Credit, Debit values
50.	7019	406	Previous IRN checker	Provide a correct previous IRN
51.	7018	406	Get Tax Center	Provide a valid tax center
52.	7017	406	Seller Details Checker	Provide a correct seller detail
53.	7016	406	Region Code for Seller	Provide a correct region code
54.	7015	406	Invoice Counter Validator	Provide a valid Invoice counter value based on previously registered invoice
55.	7014	406	Item code validator	Provide Item code if invoice is not manual type
56.	7013	406	Validate Source System	Provide a correct and registered source system

57.	7012	406	Tax Account With Withholding Validator	Provide a correct withholding amount
58.	7011	406	Currency Validator	Provide a correct and registered currency
59.	7010	406	Cancellation Reason Code	Provide a correct and registered reason code
60.	7009	406	Bulk Cancellation Limit	Please raise support ticket with MoR to raise Bulk cancellation limit.
61.	7008	406	Buyer TIN Validator	Provide a correct Buyer tin
62.	7006	406	Buyer Legal Name Validator	Provide a correct legal name
63.	7005	406	Region Code for Buyer	Provide a correct and registered region code
64.	7004	406	Buyer Id Type Validator	Provide a correct and registered buyer id
65.	7003	406	Bulk Cancellation Reason Code	Provide a correct and registered reason code
66.	7002	406	Single Cancellation Limit	Please raise support ticket with MoR to raise the single cancellation limit.
67.	7001	406	Document Number Uniqueness Checker	Provide a correct and unique document number
68.	70031	406	Validate taxpayer tax accounts (checks if the supplier is registered for the given tax accounts)	Please make sure you are using the correct tax code based on your registered tax account types

2.1.7 SUMMARY

For the taxpayer system and the Electronic Invoice Management System (EIMS) to integrate and function effectively, several key factors must be carefully addressed and prioritized. These considerations are crucial for ensuring smooth interaction between the two systems, enhancing overall efficiency, and minimizing potential challenges during the integration process. By focusing on these areas, a stable and reliable connection between the taxpayer's system and EIMS can be maintained, ensuring that all operations are executed correctly and securely.

The taxpayer system must store invoice details, including the received IRN and QR Code from EIMS, for each invoice or receipt. Tax auditors and/or inspectors will require the taxpayer to provide a list of invoices and receipts, along with the corresponding IRNs received.

- **Request Structure:** Must match the defined JSON schema, be signed, and include a valid certificate.
- **Signature & Certificate:** Digital signature required to ensure authenticity and integrity.
- **Validation Flow:**
 1. Schema Validation (if fails → 400).
 2. Business/Rule Validation (if fails → 406).
 3. On success → 200 (with IRN, QR Code and signed invoice details).

By following the guidelines in this integration compliance document, implementers can make sure their requests to the Electronic Invoicing System are well-organized, correctly signed, and processed properly. This helps ensure a smooth, error-free invoice registration process, making interactions with the system more reliable and secure. Adhering to these standards not only improves efficiency but also ensures compliance with the required regulations, resulting in a seamless integration experience.

2.1.8 ATTACHMENTS:

i. Invoice registration request details

SN	Json Path	Description	Data Type	Size/Max	Remarks	Requirement	Print
1.	"TransactionType"	Transaction Type	Select from List	3	The nature of transaction: - B2B – Business to Business - B2C – Business to Consumer/non-registered - B2G – Business to Government - G2B – Government to Business - G2C – Government to Consumer/non-registered	Mandatory	Y
2.	"DocumentDetails"."Type"	Document Type	Select from List	3	The type of invoice issued by the seller:	Mandatory	Y
3.	"DocumentDetails.ManualInvoiceNumber"	Manual Receipt Number	String	15		Conditional	Y
4.	"DocumentDetails"."DocumentNumber"	Document Number	String /accept only Numeric	100	The sequential number of invoices, debit notes and credit notes as given by the seller.	Mandatory	Y
5.	"SourceSystem"."SystemType"	Source System Type	Select from List		☐	Mandatory	
6.	"SourceSystem"."SystemNumber"	Source system Number	String	10			Y
7.	"SourceSystem"."InvoiceCounter"	Invoice counter	Numeric	20	The total number of invoices generated bythesystem or machine.	Mandatory	N
8.	"ReferenceDetail.RelatedDocument"	Related Document Number	String	100	States the related document number for which this invoice is applied or associated. Example:	Conditional	Y

					☐ Original invoice document number for which a Credit Note or Debit Note is issued. ☐ The main/first invoice number for which interim or final invoices are issued.		
9.	"ReferenceDetail.PreviousIrn"	Previous invoice IRN	String	64	The IRN associated with the last invoice generated and registered just before this one.	Mandatory	N
10.	"DocumentDetails"."Date"	Invoice Date	Date Time	-	The date and time when the e-invoice was created by the supplier. ISO 8601 standard date and time format shall be used. The default time zone is GMT+3. Format is yyyy-mm-dd hh:mm:ss.fff+/-hh:mm	Mandatory	Y
11.	"ValueDetails"."InvoiceCurrency"	Invoice Currency	Select from List	3	The currency in which the invoice is issued. The currency value shall compliant with the ISO 4217 three-character codes. This does not necessarily mean the invoice will be received in foreign currency without conversion to local currency. The value should be the currency which is legally agreed by the buyer and the seller at the time of billing.	Mandatory	Y
12.	"ValueDetails"."ExchangeRate"	Exchange Rate	Decimal	20	The rate used to convert the invoice amount from the invoice currency in to Ethiopian Birr(ETB). Required if the invoice currency is not ETB.	Conditional	Y
13.	"ValueDetails"."Item"."ExciseTaxValue"	Excise Tax Amount	Decimal	20,2	The total excise tax amount allocated for the sold goods/services. Required if the seller is obliged based on its profile.	Conditional	Y
14.		Transaction Tax Amount	Decimal	20,2	The total tax (VAT, TOT) amount collected for the sale of goods/services.	Mandatory	Y
15.	"ValueDetails"."TransactionWithholdValue"	Transaction WHT Amount	Decimal	20,2	The Actual/Expected total transaction withholding tax amount. Required if the buyer is mandated to withhold payments.	Conditional	Y

					It is actual if the payment term is immediate.		
16.	"ValueDetails"."Income WithholdValue"	Income Tax WHT Amount	Decimal		The total actual/expected direct tax amount withheld by the buyer as collection of profit/income tax in advance from the sale of goods/services. Required if the buyer is mandated to withhold payments.	Conditional	Y
17.	"ItemList"."Item"."Discount"	Total Rebate Amount	Decimal	20,2	Trade discount or rebate amount, if a trade discount on the list price is provided.	Optional	N
18.	"ValueDetails"."TotalValue"	Invoice Total Amount	Decimal	20,2	The total/gross amount inclusive of transaction taxes for the sale of goods/services. Total of the net amount and tax amounts.	Mandatory	Y
19.	"PaymentDetails"."Payment Term"	Payment Term	String		The agreed payment terms. The vendor shall provide mechanism to configure and use different terms such as “days from invoice date”, “day of month”, “end of month”, “Payment in Advance”, “Cash on delivery” and other terms as per the business practice in Ethiopia. IMMEDIATE – default and immediate payment is going to happen. Example of “days from invoice date” is %/days, %/days, ..., N/days which means 10/15, 5/20, N/30 means the buyer can get 10% discount for payments in 15 days (for “10/15”), 5% discount for payments between 16th and 20th days (for “5/20”). If the buyer does not take discounts offered, full invoice amount payment is expected by the 30th day (for “N/30”).	Mandatory	Y
20.	"PaymentDetails"."Mode"	Mode	String		Mode of Payment: Cash, Credit, Direct Transfer	Optional	Y
21.	"DocumentDetails"."Reason"	Reason	String	200	The reason for issuance of credit note, debit note, or any related transaction.	Conditional	Y

					Required if there is “Related Invoice Number” or this invoice reduces/increase an amount expected from related invoice.		
22.	"ReferenceDetails"."PurchaseOrder"	Purchase order	String	100	Purchase order number	Optional	Y
23.	"ReferenceDetails"."Contract"	Contract number	String	100	Contract number	Optional	N
24.	“SourceSystem”.”CashierName”	Cashier Name	String	100	The name of the cashier operating the EBS/who created the invoice.	Optional	Y
25.	“SourceSystem”.”SalesPersonName”	Sales Person Name	String	100	The name of the sales person/employee supplying the goods/services.	Optional	N
26.	"ReferenceDetails"."FirstTicket"	First Ticket Number	String	20	First document number of tickets or similar source numbers.	Optional	N
27.	"ReferenceDetails"."LastTicket"	Last Ticket Number	String	20	Last document number of tickets or similar source numbers.	Optional	N
SELLER DETAILS							
28.	SellerDetails		Object			Mandatory	Y
29.	"SellerDetails"."LegalName"	Legal name of seller	String	100	The registered name of the legal person supplying the goods/services.	Mandatory	Y
30.	"SellerDetails"."TIN"	Seller TIN	String	10	The Taxpayer Identification of the legal person supplying the goods/services.	Mandatory	Y
31.	"SellerDetails"."Email"	Seller Email	String	100	Email Address of Taxpayer		Y
32.	"SellerDetails"."Phone"	Seller Phone/mobile number	String	20	Mobile/Phone number of Taxpayer		Y
33.	"SellerDetails"."TradeName"	Seller Trade Name	String		Trade name is required if Subtin is provided	Conditional	Y

34.	"SellerDetails"."SubTin"	Seller Sub-TIN	String	15	The Sub-TIN of the legal person supplying the goods/services. This number is used to file returns to nearest tax center by taxpayers doing business in different distant locations. Required based on the location of the transaction and taxpayer profile.	Conditional	Y
35.	"SellerDetails"."VatNumber"	VAT Registration Number	String	13	The VAT number of the person supplying the goods/services. Required if the taxpayer is registered for VAT.	Conditional	Y
36.	"SellerDetails"."Country"	Seller Country	String			Optional	Y
37.	"SellerDetails"."Ciry"	Seller City	String	50		Required	Y
38.	"SellerDetails"."Zone"	Seller Zone	String			Optional	Y
39.	"SellerDetails"."SubCity"	Seller Subcity	String	100		Optional	Y
40.	"SellerDetails"."Region"	Seller Region	String	3		Required	Y
41.	"SellerDetails"."HouseNumber"	Seller House Number	String	50		Optional	Y
42.	"SellerDetails"."Kebele"	Seller Kebele	String			Optional	Y
43.	"SellerDetails"."Wereda"	Seller Wereda	String	50		Required	Y
44.	"SellerDetails"."Locality"	Seller Locality	String			Optional	Y
BUYER DETAILS							
45.	BuyerDetails		Object			Mandatory	Y

46.	"BuyerDetails"."LegalName"	Legal name of buyer	String	100	The legal name of the person to whom the goods / services are supplied. Required if there is "Buyer TIN".	Conditional	Y
47.	"BuyerDetails"."Tin"	Buyer TIN	String	10	The Taxpayer Identification Number of the legal person buying the goods/services. Required if the transaction is not B2C or G2C.	Conditional	Y
48.	"BuyerDetails"."SubTin"	Buyer Sub-TIN	String	15	The Sub-TIN of the legal person buying the goods/services. This number is used to file returns to nearest tax center by taxpayers doing business in different distant locations.	Conditional	Y
49.	"BuyerDetails"."VatNumber"	VAT Number - buyer	String	13	The VAT number of the person to whom the goods/services are supplied. Required if there is Buyer TIN and the buyer profile has VAT registration.	Conditional	Y
50.	"BuyerDetails"."Email"	Seller Email	String	100	Email Address of Taxpayer		Y
51.	"BuyerDetails"."Phone"	Seller Phone/mobile number	String	20	Mobile/Phone number of Taxpayer		Y
52.	"BuyerDetails"."TradeName"	Seller Trade Name	String	100	Trade name is required if Subtin is provided	Conditional	Y
53.	"BuyerDetails"."SubTin"	Seller Sub-TIN	String	15	The Sub-TIN of the legal person supplying the goods/services. This number is used to file returns to nearest tax center by taxpayers doing business in different distant locations. Required based on the location of the transaction and taxpayer profile.	Conditional	Y
54.	"BuyerDetails"."VatNumber"	VAT Registration Number	String	13	The VAT number of the person supplying the goods/services. Required if the taxpayer is registered for VAT.	Conditional	Y
55.	"BuyerDetails"."Country"	Buyer Country	String			Optional	Y

56.	"BuyerDetails"."City"	Buyer City	String	100		Required	Y
57.	"BuyerDetails"."Zone"	Buyer Zone	String			Optional	Y
58.	"BuyerDetails"."SubCity"	Buyer Subcity	String			Optional	Y
59.	"BuyerDetails"."Region"	Buyer Region	String	3		Required	Y
60.	"BuyerDetails"."HouseNumber"	Buyer House Number	String	50		Optional	Y
61.	"BuyerDetails"."Kebele"	Buyer Kebele	String	100		Optional	Y
62.	"BuyerDetails"."Wereda"	Buyer Wereda	String			Required	Y
63.	"BuyerDetails"."Locality"	Buyer Locality	String			Optional	Y
	Item						
64.	"ItemList"		Array	999			
65.	"ItemList"[index]. "Line Number"	Line Number	Numeric	4	The line-item number as given by the seller.	Mandatory	Y
66.	"ItemList"[index]. "NatureOfSupply"	Nature of Supplies	Select from List	8	The nature of supplies made to the buyer:	Mandatory	Y
67.	"ItemList"[index]. "HarmonizationCode"	Item Code by MoR	Select from List	15	Harmonized item number or HSN Code of the product/service as per the MoR specification. It can be required for specific tax payers' category or for all in future.	Conditional	Y
68.	"ItemList"[index]. "Item Code"	Item Code by Taxpayer	String	15	Item number of the product/service as per supplier's own records. Mandatory if the Source Type is not manual.	Conditional	Y

69.	"ItemList"[index]. "ProductDescription"	Description of Goods/Services	String	100	The description of goods/services supplied.	Mandatory	Y
70.	"ItemList"[index]. "UnitPrice"	Unit Price	Decimal	20,2	The unit price of goods or service supplied.	Mandatory	Y
71.	"ItemList"[index]. "Quantity"	Quantity	Numeric	20,4	The quantity of goods supplied.	Mandatory	Y
72.	"ItemList"[index]. "UnitMeasure"	Unit of Measure	Select from list	20	Short name for the unit of measure used for the quantity.	Mandatory	Y
73.	"ItemList"[index]. "RebateAmount"	Rebate amount	Decimal	20,2	Trade Discount or rebate amount, if discount is provided at line-item level.	Optional	N
74.	"ItemList"[index]. "TaxCode"	Transaction Tax Code	Select from List	5	Configurable attribute to hold transaction tax code applicable to the supplies made by the seller:	Mandatory	Y
75.	"ItemList"[index]. "PreTaxValue"	Line Amount before Tax	Decimal	20,2	The amount without any tax (transaction/excise) for the sale of goods/services expressed in the invoice currency.	Mandatory	Y
76.	"ItemList"[index]. "ExciseTaxAmount"	Excise Tax Amount	Decimal	20,2	Required if the product or service is subject to excise tax.	Conditional	Y
77.	"ItemList"[index]. "TaxAmount"	Transaction Tax Amount	Decimal	20,2	The line transaction Tax (VAT, TOT...) amount collected on the taxable line item.	Mandatory	Y
78.	"ItemList"[index]. "TotalLineAmount"	Total Line Amount	Decimal	20,2	The total amount which is sum of taxes and the line amount before tax.	Mandatory	Y

3 SECTION THREE

3.1 INVOICE REGISTRATION SCHEMA DETAIL

3.1.1 PURPOSE

For the Electronic Invoice Management System (EIMS) and the taxpayer system to work seamlessly and exchange information effectively, it's crucial to follow a predefined JSON schema. This integration document is designed to guide technical teams in connecting their systems to the e-invoicing platform using the provided schema. It explains how to correctly structure invoice requests, understand the necessary conditions, and validate data before submitting it. By following these instructions, can ensure smooth communication between taxpayer systems and EIMS, helping maintain accuracy and reliability in the invoicing process.

3.1.2 OVERVIEW OF THE E-INVOICING SCHEMA

The provided schema helps ensure that data remains consistent, accurate, and compliant for electronic invoices, credit notes, debit notes, and other related documents. By following this schema, taxpayer can stay in line with local e-invoicing regulations, reducing the chances of errors and ensuring that all documents meet the required standards. This leads to a more efficient invoicing process and reduces compliance risks, making the entire system more reliable and easier to work with.

3.1.3 HIGH-LEVEL ARCHITECTURE

3.1.3.1 SYSTEM COMPONENTS

- **Your ERP/CRM/POS:** Generates invoice data.
- **E-Invoicing Validation Service:** Applies the JSON schema rules.
- **E-Invoicing:** Receives and processes validated invoices.

3.1.3.2 DATA FLOW

1. Your system creates the invoice JSON object.
2. Validate the object against the schema.
3. Submit the validated invoice to the EIMS API.
4. Receive an Invoice Reference Number (IRN) and QR Code or error messages.
5. Store the IRN and invoice data.

3.1.4 SCHEMA OVERVIEW

3.1.4.1 JSON Schema Version

The schema is based on JSON Schema draft-07, so it's crucial to ensure that your validator supports this version. This will help guarantee that the data is properly validated and meets the required standards. Using the correct version of the schema will prevent potential issues and ensure everything functions smoothly.

3.1.4.2 Top-Level Elements

Key properties include:

- **TransactionType**
- **DocumentDetails**
- **SourceSystem**
- **SellerDetails**
- **BuyerDetails**
- **ItemList**
- **PaymentDetails**
- **ValueDetails**
- **ReferenceDetails**

All these have specific constraints and sometimes conditional requirements.

3.1.4.3 Key Entities and Their Properties

TransactionType

- **Description** : Indicates the type of transaction (e.g., B2B, B2C, G2C).
- **Purpose** : To classify the nature of the transaction between the seller and buyer.
- **Format** : String with a maximum length of 3 characters.
- **Example Values** : "B2B", "B2C", "B2G", "G2B", "G2C", "
- **Conditional Validations** :
 - If TransactionType is not "B2C" or "G2C", then BuyerDetails.Tin is required and must be a non-null string.
 - If TransactionType is "B2C" or "G2C", then BuyerDetails.Tin can be null, and BuyerDetails.VatNumber must also be null.

DocumentDetails

- **Description** : Contains metadata about the document, such as its type, number, date, and reason (if applicable).

Sub-Fields

- **Type**
 - **Purpose** : Specifies the type of document (Invoice (INV), Credit Note (CRE), Debit (DEB) Note, Interim (INT), Retainer (RTN), Final (FIN), Mixed Invoice (MIX), Intercompany (INC), Proforma (PRF), Overdue (OVD)).
 - **Format** : A string with a maximum length of 3 characters. Example: "INV", "CRE", "DEB".
 - **Conditional Validation** : If Type is "CRE", "DEB", "INT", or "FIN", then Reason is required.
- **DocumentNumber**
 - **Purpose** : Sequential number assigned by the seller to the document.
 - **Format** : A string with a length between 1 and 100 characters. Example: "1234".
- **Date**
 - **Purpose** : Date and time when the document was generated.
 - **Format** : ISO 8601 datetime string. Example: "15-09-2023T14:30:00".
- **Reason**
 - **Purpose** : Reason for issuing the document (applicable for credit/debit notes).
 - **Format** : A string with a length between 3 and 100 characters. Example: "Test reason".
 - **Conditional Validation** : Required if Type is "CRE", "DEB", "INT", or "FIN".

SourceSystem

- **Description** : Identifies the system that generated the invoice.

Sub-Fields

- **SystemType**
 - **Purpose** : Type of source system (ERP, Manual, Cash Register Machine, etc.).
 - **Format** : A string with a length of exactly 3 characters. Example: "SYS", "MAN", "CRM".
- **CashierName**
 - **Purpose** : Name of the cashier operating the system.
 - **Format** : A string with a length between 3 and 100 characters. Example: "John Doe".
 - **Optional** .
- **SystemNumber**
 - **Purpose** : Identifier for the system (e.g., cash register machine number).
 - **Format** : A string with a maximum length of 10 characters. Example: "MRC12345".
 - **Optional** .
- **InvoiceCounter**
 - **Purpose** : Total number of invoices generated by the system.
 - **Format** : An integer greater than or equal to 1. Example: 12345.
 - **Optional** .
- **SalesPersonName**
 - **Purpose** : Name of the salesperson responsible for the transaction.
 - **Format** : A string with a length between 3 and 100 characters. Example: "Jane Smith".
 - **Optional** .

SellerDetails

❖ **Description** : Provides detailed information about the seller.

Sub-Fields

- **Tin**
 - **Purpose** : Tax Identification Number of the seller.
 - **Format** : A string with a length between 10 and 20 digits. Example: "1234567890".
 - **Mandatory** .
- **LegalName**
 - **Purpose** : Legal name of the seller.
 - **Format** : A string with a length between 3 and 100 characters. Example: "ABC Corporation".
 - **Mandatory** .
- **City**
 - **Purpose** : City where the seller operates.
 - **Format** : A string with a length between 1 and 100 characters. Example: "Addis Ababa".
 - **Mandatory** .
- **Wereda**
 - **Purpose** : Administrative subdivision (Woreda) of the seller's location.

- **Format** : A string with a length between 1 and 50 characters. Example: "Arada".
- **Mandatory** .
- **Region**
 - **Purpose** : Region code where the seller operates.
 - **Format** : A string with a length between 1 and 3 digits. Example: "11".
 - **Mandatory** .
- **Zone**
 - **Purpose** : Zone code of the seller's location.
 - **Format** : A string with an optional value. Example: "Zone 1".
 - **Optional** .
- **Email**
 - **Purpose** : Email address of the seller.
 - **Format** : Valid email format. Example: "info@abccorp.com".
 - **Optional** .
- **Phone**
 - **Purpose** : Phone number of the seller.
 - **Format** : A string with a length between 6 and 20 characters. Example: "0112345678".
 - **Optional** .
- **Kebele**
 - **Purpose** : Kebele code of the seller's location.
 - **Format** : A string with an optional value. Example: "Kebele 01".
 - **Optional** .
- **SubTin**
 - **Purpose** : Sub-TIN for taxpayers operating in multiple locations.
 - **Format** : A string with a length between 10 and 25 characters. Example: "1234567890-01".
 - **Optional** .
- **Country**
 - **Purpose** : Country code of the seller's location.
 - **Format** : A string with an optional value. Example: "ET".
 - **Optional** .
- **SubCity**
 - **Purpose** : Sub-city of the seller's location.
 - **Format** : A string with an optional value. Example: "Bole".
 - **Optional** .
- **Locality**
 - **Purpose** : Locality of the seller's location.
 - **Format** : A string with a length between 3 and 50 characters. Example: "Kirkos".
 - **Optional** .
- **TradeName**
 - **Purpose** : Trade name of the seller (if different from legal name).
 - **Format** : A string with a length between 3 and 100 characters. Example: "ABC Store".

- **Optional .**
- **VatNumber**
 - **Purpose** : VAT registration number of the seller.
 - **Format** : A string with a length between 3 and 25 digits. Example: "123456789".
 - **Conditional Validation** : Required if any TaxCode in ItemList starts with "VAT".
- **HouseNumber**
 - **Purpose** : House number of the seller's address.
 - **Format** : A string with a length between 3 and 50 characters. Example: "House 123".
 - **Optional .**

BuyerDetails

❖ **Description** : Provides detailed information about the buyer.

Sub-Fields

- **LegalName**
 - **Purpose** : Legal name of the buyer.
 - **Format** : A string with a length between 3 and 100 characters. Example: "XYZ Ltd".
 - **Mandatory .**
- **Tin**
 - **Purpose** : Tax Identification Number of the buyer.
 - **Format** : A string with a length between 10 and 20 digits. Example: "9876543210".
 - **Conditional Validation** : Required if TransactionType is not "B2C" or "G2C". Must be null if TransactionType is "B2C" or "G2C".
- **City**
 - **Purpose** : City where the buyer operates.
 - **Format** : A string with a length between 1 and 100 characters. Example: "Addis Ababa".
 - **Optional .**
- **Zone**
 - **Purpose** : Zone code of the buyer's location.
 - **Format** : A string with an optional value. Example: "Zone 1".
 - **Optional .**
- **Email**
 - **Purpose** : Email address of the buyer.
 - **Format** : Valid email format. Example: "contact@xyz.com".
 - **Optional .**
- **Phone**
 - **Purpose** : Phone number of the buyer.
 - **Format** : A string with a length between 6 and 20 characters. Example: "0119876543".
 - **Optional .**
- **IdType**
 - **Purpose** : Type of identification provided by the buyer.

- **Format** : A string with a length of exactly 3 characters. Example: "NID".
 - **Optional** .
- **Kebele**
 - **Purpose** : Kebele code of the buyer's location.
 - **Format** : A string with an optional value. Example: "Kebele 01".
 - **Optional** .
- **Region**
 - **Purpose** : Region code where the buyer operates.
 - **Format** : A string with a length between 1 and 3 digits. Example: "11".
 - **Optional** .
- **SubTin**
 - **Purpose** : Sub-TIN for taxpayers operating in multiple locations.
 - **Format** : A string with a length between 10 and 25 characters. Example: "9876543210-01".
 - **Optional** .
- **Wereda**
 - **Purpose** : Administrative subdivision (Woreda) of the buyer's location.
 - **Format** : A string with an optional value. Example: "Arada".
 - **Optional** .
- **Country**
 - **Purpose** : Country code of the buyer's location.
 - **Format** : A string with an optional value. Example: "ET".
 - **Optional** .
- **SubCity**
 - **Purpose** : Sub-city of the buyer's location.
 - **Format** : A string with an optional value. Example: "Bole".
 - **Optional** .
- **IdNumber**
 - **Purpose** : Unique identification number of the buyer (e.g., National ID).
 - **Format** : A string with a length between 1 and 20 characters. Example: "123456789".
 - **Optional** .
- **Locality**
 - **Purpose** : Locality of the buyer's location.
 - **Format** : A string with a length between 1 and 50 characters. Example: "Kirkos".
 - **Optional** .
- **TradeName**
 - **Purpose** : Trade name of the buyer (if different from legal name).
 - **Format** : A string with a length between 3 and 100 characters. Example: "XYZ Shop".
 - **Optional** .
- **VatNumber**
 - **Purpose** : VAT registration number of the buyer.
 - **Format** : A string with a length between 3 and 25 digits. Example: "987654321".

- **Conditional Validation** : Must be null if Tin is null.
- **HouseNumber**
 - **Purpose** : House number of the buyer's address.
 - **Format** : A string with a length between 3 and 50 characters. Example: "House 456".
 - **Optional** .

ItemList

- ❖ **Description** : Lists all items included in the transaction.

Sub-Fields

- **LineNumber**
 - **Purpose** : Line-item number assigned by the seller.
 - **Format** : An integer between 0 and 99. Example: 1.
 - **Mandatory** .
- **NatureOfSupplies**
 - **Purpose** : Description of the goods/services supplied.
 - **Format** : A string with a length between 1 and 300 characters. Example: "Laptop - Model X1".
 - **Mandatory** .
- **UnitPrice**
 - **Purpose** : Price per unit of the item.
 - **Format** : A number greater than or equal to 0. Example: 1000.00.
 - **Mandatory** .
- **TotalLineAmount**
 - **Purpose** : Total amount for the line item (including taxes).
 - **Format** : A number greater than or equal to 0, with a maximum value of 999999999999.99. Example: 1100.00.
 - **Mandatory** .
- **PreTaxValue**
 - **Purpose** : Value of the item before tax.
 - **Format** : A number greater than or equal to 0. Example: 1000.00.
 - **Mandatory** .
- **Unit**
 - **Purpose** : Unit of measure for the quantity.
 - **Format** : A string with a length between 3 and 8 characters. Example: "PCS".
 - **Mandatory** .
- **TaxCode**
 - **Purpose** : Code representing the type of tax applied.
 - **Format** : A string with a length between 3 and 5 characters. Example: "VAT".
 - **Mandatory** .
- **TaxAmount**
 - **Purpose** : Tax amount for the line item.

- **Format** : A number greater than or equal to 0. Example: 100.00.
 - **Mandatory** .
- **Quantity**
 - **Purpose** : Quantity of the item sold.
 - **Format** : A number greater than or equal to 0. Example: 1.
 - **Mandatory** .
- **Discount**
 - **Purpose** : Discount amount applied to the line item.
 - **Format** : A number greater than or equal to 0. Example: 50.00.
 - **Optional** .
- **HarmonizationCode**
 - **Purpose** : Harmonized System (HS) code for the item.
 - **Format** : A string with a length of 4, 6, or 8 digits. Example: "8517".
 - **Optional** .
- **ProductDescription**
 - **Purpose** : Detailed description of the product/service.
 - **Format** : A string with a length between 3 and 300 characters. Example: "High-performance laptop".
 - **Optional** .
- **ItemCode**
 - **Purpose** : Internal code for the item in the seller's records.
 - **Format** : A string with a length between 1 and 15 characters. Example: "PROD123".
 - **Conditional Validation** : Required if SourceSystem.SystemType is not "MAN".

ValueDetails

❖ **Description** : Captures financial details of the transaction.

Sub-Fields

- **TotalValue**
 - **Purpose** : Total value of the transaction (inclusive of taxes).
 - **Format** : A number greater than or equal to 0. Example: 1100.00.
 - **Mandatory** .
- **TaxValue**
 - **Purpose** : Total tax amount collected for the transaction.
 - **Format** : A number greater than or equal to 0. Example: 100.00.
 - **Mandatory** .
- **Discount**
 - **Purpose** : Overall discount applied to the transaction.
 - **Format** : A number greater than or equal to 0. Example: 50.00.
 - **Optional** .
- **ExciseValue**
 - **Purpose** : Excise tax amount (if applicable).

- **Format** : A number greater than or equal to 0. Example: 20.00.
- **Conditional Validation** : Required if the seller is obliged based on its profile.
- **ExchangeRate**
 - **Purpose** : Exchange rate used to convert the invoice currency to Ethiopian Birr (ETB).
 - **Format** : A number greater than or equal to 1. Example: 30.00.
 - **Conditional Validation** : Required if InvoiceCurrency is not "ETB".
- **InvoiceCurrency**
 - **Purpose** : Currency in which the invoice is issued.
 - **Format** : A string with a length of exactly 3 characters. Example: "USD".
 - **Conditional Validation** : If not "ETB", then ExchangeRate is required.
- **IncomeWithholdValue**
 - **Purpose** : Income tax withheld by the buyer.
 - **Format** : A number greater than or equal to 0. Example: 50.00.
 - **Conditional Validation** : Required if the buyer is mandated to withhold payments.
- **TransactionWithholdValue**
 - **Purpose** : Transaction tax withheld by the buyer.
 - **Format** : A number greater than or equal to 0. Example: 30.00.
 - **Conditional Validation** : Required if the buyer is mandated to withhold payments.

PaymentDetails

❖ *Description* : Specifies payment-related information.

Sub-Fields

- **PaymentTerm**
 - **Purpose** : Agreed terms for payment (e.g., Cash, Advance, After n days).
 - **Format** : A string with a length between 3 and 100 characters. Example: "CASH", "IMMEDIATE".
 - **Mandatory** .
- **Mode**
 - **Purpose** : Mode of payment (e.g., Cash, Credit, Direct Transfer).
 - **Format** : A string with a length between 1 and 18 characters. Example: "Cash", "Credit".
 - **Optional** .
- **Conditional Validation** : If PaymentTerm is not "IMMEDIATE", then TransactionWithholdValue or IncomeWithholdValue in ValueDetails are required.

ReferenceDetails

❖ *Description* : Links the current document to related documents.

Sub-Fields

- **RelatedDocument**
 - **Purpose** : Identifier for the related document (e.g., original invoice for credit/debit notes).
 - **Format** : A string. Example: "INV12345".

- **Conditional Validation** : Required if DocumentDetails.Type is "CRE", "DEB", "INT" or "FIN".

3.1.4.4 Conditional and Contextual Validation Rules

- **Credit/Debit Notes**

If *DocumentDetails.Type* is CRE or DEB:

- Reason must be provided in *DocumentDetails*.
- RelatedDocument must be provided in *ReferenceDetails*.

- **Foreign Currency Invoices**

If *ValueDetails.InvoiceCurrency* ≠ "ETB":

- ExchangeRate is required.

- **VAT Items**

If any item in *ItemList* has *TaxCode* = "VAT"(prefix VAT eg: VAT15, VAT0, VATEX. This can be configured on the Backoffice):

- SellerDetails.VatNumber is required.

- **Buyer TIN Length**

If *BuyerDetails.Tin* is filled:

- BuyerDetails.LegalName is required.

- **TransactionType and Buyer TIN**

If *TransactionType* is not B2C or G2C:

- BuyerDetails.Tin is required and must be a string. Otherwise, Tin can be string or null.

- **Null Buyer TIN**

If *BuyerDetails.Tin* is null:

- BuyerDetails.VatNumber must be null.
- Sub-TIN...

3.1.4.5 Required Fields and Validation Constraints

- **Global Required Fields**

At the top level, the following fields must always appear:

- TransactionType
- DocumentDetails
- SourceSystem
- SellerDetails
- BuyerDetails
- ItemList
- PaymentDetails
- ReferenceDetails

- **Field Length, Patterns, And Data Types**

- *Many string fields have min/max lengths and regex patterns .*
- *Numerical fields have defined min/max values.*
- *Date fields must follow yyyy-mm-dd hh:mm:ss.fff+/-hh:mm.*

- **Numeric Limits**

Use appropriate data types to handle large numeric fields without overflow.

3.1.4.6 Integration Steps

- **Preparing The Invoice Data**

Assemble a JSON object meeting all schema requirements.

Gather required master data (e.g., valid Region codes, Seller TIN).

- **Validating Against The Schema**

Use a JSON Schema validator to ensure compliance before sending.

- **Submitting The Invoice**

Send the validated JSON to the e-invoicing portal's endpoint.

3.1.4.7 Handling Validation Errors

If validation fails, log errors, correct data, and re-validate.

3.1.4.8 Storing and Retrieving Invoices

Store both raw JSON, QR Code and IRN for future reference and audits.

3.1.4.9 Error Handling and Troubleshooting

- **Common Validation Errors**

- Missing required fields (e.g., Reason for a credit note).
- Invalid patterns (e.g., TIN not matching required format).
- Missing *ExchangeRate* for foreign currency invoices.

- **Debugging Schema Violations**

Use verbose logging from the validator to pinpoint issues.

- **Recommended Logging Practices**

Log invoice data and errors with timestamps for easy traceability.

3.1.4.10 Security And Compliance

- **Data Integrity And Non-Repudiation**

Maintain the integrity of invoices and IRNs to prevent tampering.

- **Sensitive Data Considerations**

Handle TINs and VAT numbers securely and in compliance with regulations.

3.1.4.11 Examples

- **Minimal Valid Invoice (B2c)**

```
{
  "BuyerDetails": {
    "City": "0",
    "Email": "test@test.com",
    "HouseNumber": "NEW",
    "IdNumber": "111222",
    "IdType": "KID",
    "Tin": "0123456789",
    "LegalName": "TEST PLC",
    "Phone": "0912345678",
    "Region": "13",
```



```
"Country": "70",
"Zone": "SHA",
"Kebele": "WOREDA 03",
"VatNumber": "0123456789",
"Wereda": "574"
},
"DocumentDetails": {
  "DocumentNumber": "1",
  "Date": "10-02-2025T00:59:00",
  "Type": "INV"
},
"ItemList": [
  {
    "Discount": 0,
    "ExciseTaxValue": 0,
    "HarmonizationCode": "1111",
    "NatureOfSupplies": "Goods",
    "ItemCode": "1111",
    "ProductDescription": "string",
    "PreTaxValue": 1000,
    "Quantity": 1,
    "LineNumber": 1,
    "TaxAmount": 150,
    "TaxCode": "VAT15",
    "TotalLineAmount": 1150,
    "Unit": "PCS",
    "UnitPrice": 1000
  },
  {
    "Discount": 0,
    "ExciseTaxValue": 0,
    "HarmonizationCode": "1111",
    "NatureOfSupplies": "Goods",
    "ItemCode": "1111",
    "ProductDescription": "Phone",
    "PreTaxValue": 10000,
    "Quantity": 1,
    "LineNumber": 2,
    "TaxAmount": 1500,
    "TaxCode": "VAT15",
    "TotalLineAmount": 11500,
    "Unit": "PCS",
    "UnitPrice": 10000
  }
],
"PaymentDetails": {
```

```

    "Mode": "CASH",
    "PaymentTerm": "IMMEDIATE"
  },
  "ReferenceDetails": {
    "PreviousIrn":
"635d66bdce10cf8ff34b9ea5729f22354f824a9b39f16bcd6f6ccea870505f94a"
  },
  "SellerDetails": {
    "City": "0",
    "Email": "test2@test2.com",
    "HouseNumber": "new",
    "LegalName": "TEST2 PLC",
    "Locality": "AAA",
    "Phone": "0116789",
    "Region": "3",
    "SubCity": "AAA",
    "Tin": "0023456789",
    "VatNumber": "0113456789",
    "Wereda": "883",
    "Country": "70"
  },
  "SourceSystem": {
    "InvoiceCounter": 1,
    "SystemNumber": "650363DBE3",
    "SystemType": "POS"
  },
  "TransactionType": "B2C",
  "ValueDetails": {
    "Discount": null,
    "ExciseValue": 0,
    "IncomeWithholdValue": 220,
    "TaxValue": 1650,
    "TotalValue": 12650,
    "TransactionWithholdValue": 0,
    "InvoiceCurrency": "ETB"
  }
}

```

- **Foreign Currency Example**

- Include "InvoiceCurrency": "USD" and specify "ExchangeRate": 50.00" in ValueDetails.

- **Credit Note Example**

- Set *DocumentDetails.Type* = "CRE"
- Add "Reason": "Return of goods" in *DocumentDetails*
- Provide *RelatedDocument* in *ReferenceDetails*.

3.1.4.12 Appendices

- **Change Management And Versioning**

Track schema versions as indicated in Version field and portal documentation.

- **Useful References**

- JSON Schema Draft-07
- Local E-Invoicing Portal Documentation
- ISO 4217 Currency Codes

2. SECTION FOUR

2.1. BULK INVOICE REGISTRATION

2.1.1. PURPOSE

In the Electronics Invoice Management System (EIMS), there are two main ways to register invoices: single invoice registration and bulk (or batch) invoice registration, which allows you to register multiple invoices at once. This section is here to guide through the process of bulk invoice registration, making it easier to handle large numbers of invoices efficiently.

This section of the document provides clear, step-by-step instructions on how to register invoices in bulk. It also offers helpful tips, best practices, and advice for tackling common challenges, ensuring that you can smoothly and accurately manage the bulk registration process without unnecessary issues.

2.1.2. OVERVIEW BULK INVOICE REGISTRATION

Bulk e-invoice registration is a helpful feature that allows taxpayer systems to register multiple invoices all at once. This is especially useful in situations where the system needs to gather invoice documents locally and then register them in bulk at a later time. It's designed to make the process more efficient, particularly when dealing with large numbers of invoices.

This feature is perfect for businesses that need to register invoices in batches, whether they've accumulated a group of invoices over time or need to process a large volume all at once. Below, we'll cover some important things to keep in mind when using bulk invoice registration to ensure everything goes smoothly.

1. The number of invoice documents that can be sent for bulk registration at once is governed by a rule set
2. Taxpayer systems when sending bulk invoice registration request will receive conversation id if all the invoice documents in the request comply with the schema standard of EIMS electronic invoice registration

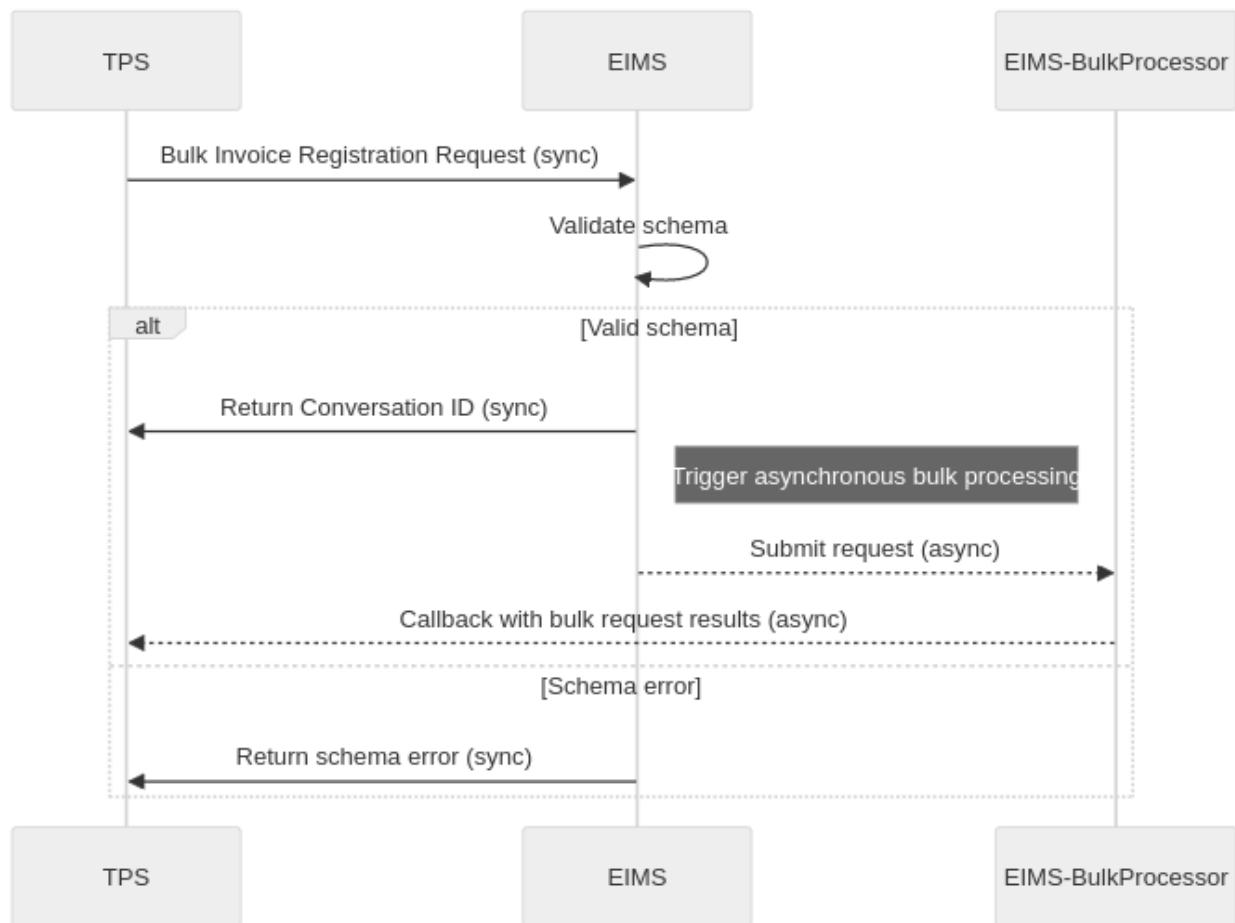
3. EIMS will process bulk invoice registration asynchronously and use a call back url and provided conversation id to notify TP systems of the invoice registration result. In case of failed call back handling or if the sales register machine cannot receive call back from EIMS, it can consume the API endpoint availed by EIMS to fetch status and response contents by conversation ID.
4. Invoice registration failures related to rule evaluation will be handled separately for each invoice and appropriate responses related to each invoice will be part of the bulk invoice registration response

2.1.3. HIGH-LEVEL ARCHITECTURE

2.1.3.1. System Components

- **Your ERP,CRM, POS or billing system:** Generates bulk invoice registration data.
- **E-Invoicing Validation Service:** Applies the JSON schema rules and if the rules are satisfied a conversation id will be returned.
- **EIMS backend:** Receives and processes validated invoices, store the results and call the provided call back url to provide the result
- **Taxpayer System:** will receive the results and process.

The taxpayer must retain the received IRN and QR Code for each invoice or receipt. Tax auditors and/or supervisors will request the taxpayer to provide the list of invoices and receipts against the list of received IRNs.



2.1.3.2. Data Flow

1. Taxpayer system creates the bulk invoice JSON object.
2. EIMS validate the object against the schema.
3. EIMS will send conversation id for asynchronous communication reference, if the schema validation passes.
4. Taxpayer system will receive results of bulk registration through call back URL.

2.1.4. BULK INVOICE REGISTRATION REQUEST

2.1.4.1. End-Point Details

- **HTTP Method:** POST
- **Endpoint URL:** <https://eims.mor.gov.et/api/v1/bulkInvoice>

2.1.4.2. Headers

The following headers must be included in every request:

Header Key	Description	Example Value
TIN	Taxpayer Identification Number of the seller	
callback	The URL through which a call back will be made when results are ready	https://tpsystem/bulkcallback
system_number	EIMS assigned system number	

Figure 5: bulk upload header

2.1.4.3. Request Format

The request body must be a JSON array, where each element is an **invoice-document** in JSON format.

Each **invoice-document** is a JSON object that is based on the invoice registration schema and will be validated for each invoice in the array.

The following is an example of a bulk invoice registration request using the curl utility,

```
curl --location 'http://eims.mor.gov.et/api/v1/bulkRegister' \  
--header 'Content-Type: application/json' \  
--header 'bearer: xxxxxxxxxx' \  
--data-raw '[  
  {  
    "BuyerDetails": {  
      "City": "AAB",  
      "Email": "AA@xyz.com.et",  
      "HouseNumber": "111",  
      "IdNumber": "1112222222222222",  
      "IdType": "KID",
```

```
"Tin": "1234567899",

"LegalName": "Kebeda Balcha Safo",
"Phone": "251929909090",
"Region": "AAA",
"Country": null,
"Zone": "SHA",
"Kebele": "002"
},
"DocumentDetails": {
  "DocumentNumber": "8007",
  "Date": "19-12-2024T00:55:52",

  "Type": "INV",
  "ManualInvoiceNumber": 2
},
"ItemList": [
  {
    "Discount": 0,
    "ExciseTaxValue": 10,
    "HarmonizationCode": "1111xxxx",
    "NatureOfSupplies": "Service",
    "ItemCode": "1111",
    "ProductDescription": "string",
    "PreTaxValue": 200,
    "Quantity": 2,
    "LineNumber": 1,
    "TrxnTaxAmount": 40,
    "TaxCode": "VAT15",

    "TotalLineAmount": 240,
    "Unit": "PCS",
    "UnitPrice": 100
  }
],
```



```
"PaymentDetails": {
  "Mode": "CASH",
  "PaymentTerm": "IMMEDIATE"
},
"ReferenceDetails": {
  "PreviousIrn": "4e8e6a26be7b15fa3e423f987070ca2a804e2843f2dc3c43f05097642a9971a5",
  "RelatedDocument": "1234"
},
"SellerDetails": {

  "City": "AAA",
  "Email": "AA@xyz.com",
  "HouseNumber": "AAA",
  "LegalName": "AAA",
  "Locality": "AAA",
  "Phone": "25191111111",
  "Region": "AAA",
  "SubCity": "AAA",
  "Tin": "2345678901",
  "SubTin": "2345678901-1",
  "VatNumber": "1234567890",
  "Wereda": "AAA",
},
"SourceSystem": {
  "CashierName": "AAA",
  "InvoiceCounter": 1,
  "SalesPersonName": "AAA",
  "SystemNumber": "GHEFR8ZYZO",
  "SystemType": "MAN"
},
"TransactionType": "B2C",
"ValueDetails": {
  "Discount": null,
  "ExciseValue": 10,
  "IncomeWithholdValue": 0,
```

```
"TaxValue": 40,
"TotalValue": 240,
"TransactionWithholdValue": 0,
"ExchangeRate": 122,
"InvoiceCurrency": "ETB"
},
},
{
  "BuyerDetails": {
    "City": "AAB",
    "Email": "AA@xyz.com.et",
    "HouseNumber": "111",
    "IdNumber": "1112222222222222",
    "IdType": "KID",
    "Tin": "1234567899",

    "LegalName": "Kebeda Balcha Safo",
    "Phone": "251929909090",
    "Region": "AAA",
    "Country": null,
    "Zone": "SHA",
    "Kebele": "002"
  },
  "DocumentDetails": {
    "DocumentNumber": "8008",
    "Date": "19-12-2024T00:55:52",

    "Type": "INV"
  },
  "ItemList": [
    {
      "Discount": 0,
      "ExciseTaxValue": 10,
      "HarmonizationCode": "1111",
```

```
"NatureOfSupplies": "Service",
"ItemCode": "1111",
"ProductDescription": "string",
"PreTaxValue": 200,
"Quantity": 2,
"LineNumber": 1,
"TaxAmount": 40,
"TaxCode": "VAT15",
"TotalLineAmount": 240,
"Unit": "PCS",
"UnitPrice": 100
}
],
"PaymentDetails": {
  "Mode": "CASH",
  "PaymentTerm": "IMMEDIATE"
},
"ReferenceDetails": {
  "PreviousInn": null,
  "RelatedDocument": "1234"
},
"SellerDetails": {
  "City": "AAA",
  "Email": "AA@xyz.com",
  "HouseNumber": "AAA",
  "LegalName": "AAA",
  "Locality": "AAA",
  "Phone": "251911111111",
  "Region": "AAA",
  "SubCity": "AAA",
  "Tin": "2345678901",
  "SubTin": "2345678901-1",
  "VatNumber": "1234567890",
  "Wereda": "AAA",
```

```

},
"SourceSystem": {
  "CashierName": "AAA",
  "InvoiceCounter": 1,
  "SalesPersonName": "AAA",
  "SystemNumber": "GHEFR8ZYZO",
  "SystemType": "MAN"
},
"TransactionType": "B2C",
"ValueDetails": {
  "Discount": null,
  "ExciseValue": 10,
  "IncomeWithholdValue": 0,
  "TaxValue": 40,
  "TotalValue": 240,
  "TransactionWithholdValue": 0,
  "ExchangeRate": 122,
  "InvoiceCurrency": "ETB"
},
}
]

```

2.1.4.4. Responses

As explained earlier, the bulk registration process involves two types of responses. The first response is sent immediately after you submit the initial request. This response confirms that the request has been received and is being processed. The second response, which is the final result of the bulk registration, is sent to the callback URL you've provided. This final response gives detailed information about each invoice in the batch, letting you know whether the registration was successful or if any issues came up.

The sections below will go into more detail about each type of response, explaining what they mean, how they're formatted, and how to handle them. This will help you understand the process clearly and make it easier to work with bulk registration responses.

2.1.4.4.1. Successful Synchronous Response

The Electronics Invoice Management System (EIMS) will first check all the invoices in the bulk registration request to make sure they follow the correct format and structure, as defined by the system's schema. This validation step ensures that every invoice in the batch meets the required standards. If all the invoices pass this check, EIMS will send back a response confirming that the validation was successful. This response will include details about the results, giving you the confidence that the request is properly formatted and ready to move on to the next step in the process.

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "Response Schema",
  "type": "object",
  "properties": {
    "statusCode": {
      "type": "integer",
      "description": "An integer representing the HTTP status code (e.g., 202 for Accepted)."
    },
    "conversationId": {
      "type": "string",
      "description": "Unique conversation or correlation ID."
    }
  },
  "required": [
    "statusCode",
    "conversationId"
  ]
}
```

Below is an example of the above response,

```
{
  "status": 202,
  "conversationId": "2345678901-1734863540359-1119cfdd-8f25-4bec-8fee-9d921a5abb5f"
}
```

2.1.4.4.2. Failed Asynchronous Response

If the request fails because one or more invoices in the request don't meet the required schema format, EIMS will send back a response explaining what went wrong. This response will include details about the specific invoices that didn't pass validation and the reasons for the failure. By providing this information, the system helps you quickly identify and fix the issues so you can correct the errors and resubmit the request. The structure of the response is outlined below and is designed to clearly show the validation errors for easy troubleshooting.

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "Response Schema",
  "type": "object",
  "properties": {
    "status": {
      "type": "integer",
      "description": "An integer representing the HTTP status code (usually 400).",
    },
    "message": {
      "type": "string",
      "description": "Description of the error encountered (usually SCHEMA ERROR)",
    },
    "body": {
      "type": "array",
      "description": "Array of objects containing error details in schema validation",
    },
  },
  "required": [
    "status",
    "message",
    "body"
  ]
}
```

The following is an example of schema error in bulk registration,

```

{
  "statusCode": 400,
  "message": "SCHEMA ERROR",
  "body": [
    {
      "type": "required",
      "code": "1028",
      "message": "$.DocumentDetails: required property 'Type' not found",
      "instanceLocation": "$.DocumentDetails",
      "property": "Type",
      "evaluationPath": "$.properties.DocumentDetails.required",
      "schemaLocation": "http://www.mor.gov.et/einvoice#/properties/DocumentDetails/required",
      "messageKey": "required",
      "arguments": [
        "Type"
      ]
    },
    {
      "type": "required",
      "code": "1028",
      "message": "$.DocumentDetails: required property 'Reason' not found",
      "instanceLocation": "$.DocumentDetails",
      "property": "Reason",
      "evaluationPath": "$.allOf[0].then.properties.DocumentDetails.required",
      "schemaLocation": "http://www.mor.gov.et/einvoice#/allOf/0/then/properties/DocumentDetails/required",
      "messageKey": "required",
      "arguments": [
        "Reason"
      ]
    }
  ]
}

```

2.1.4.4.3. Asynchronous Call Back Response

When the bulk registration request is processed the result will be sent to the call back URL registered during the bulk request. The schema of the result is similar to the one sent back during single invoice registration with successful responses containing IRN, Signed QR, Signed Invoice and invoice details and failed responses containing failure details including rule validation errors.

The schema of the response is as follows,

```

{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "Callback Schema",
  "type": "object",
  "properties": {
    "conversationId": {
      "type": "string",

```

```

    "description": "The conversation id assigned during the initial request"
  },
  "body": {
    "type": "array",
    "description": "Invoice registration request responses"
  }
},
"required": [
  "conversationId",
  "body"
]
}

```

The following is an example of a call back with results,

```

[
  {
    "irn": "d8f358d9e279ad8dcfba3c778cdafb6d597da53d32a79260368979e0f5ecb65d",
    "ackDate": "2024-12-20T11:51:05.639910+03:00[Africa/Addis_Ababa]",
    "signedQR":
    "iVBORw0KGgoAAAANSUHEUgAAAMgAAADIAQAAAACFI5MzAAAFf0IEQVR4Xu3SezSUaRwHcJqKdQ
    m7GRXl0pSSRi5tLvHWsZLQ0E1RaWtXFNGplbR6ZZxYOcaOEOmimhq3CEkjuUSbDKuLLiiXLmNiOi7ZvDPv
    XPY4w5zzm7N7zv6xf873z+/n+T3v+zznUZP9W9SUC0VUopKJqEQIE1HJfxBcx943uL3nwVXa/j0IEGKweWi/IP
    XjbD5hgWERWcdhb2VKGskyk9hKYpWCJWq4/+zQujxUSaxLWkiVm5vsGI5KQnwaquFoFx9JZAdFQsG1CqdyG
    QEik53xcDIxPViaPYBNFIMzFrHfnjKpG6bIvLaiQAgiakAjxti9NNBhHAEis7A0Dhrgr9P/q9Npspma+dqkufLDym
    6GJW8jFNzz+CN/U77g+5JYpduRIB0Nt9l9/sFoT+1rBAj+Z0SAwS3KzuxYWQAKhKiW0gJ4hpTQddxUDlg4gko
    2u08ML+qKWAVF8kvymNMdLD00ecxBIgwpG5L8qvYgPk6dzJQINJq/pOFT4JZV4LrpFCiIF8zlxQffzP22NtI3k
    yJqGZW2rFncSnIDWG5GBBcT11XmxX95sd3pg4oFPv8+/o4A0M8Ft7EgljThrnXWRtGbmWuT+WN4jwXkluC9
    V3OjGo4dGFAxNSYbxKPXWEtiplDgVCbFHT1HE0Yg3evuODAMFJLm1q8738yClu6lCkjrIwl+3B2uV6oY5Q
    hFaFTCt26LQMZ/0iFEqNYC4nn5FRcOejIwZESkI96bPXhJoVLKMjQE9mW/XMg113brHEqBINHVCihY1H0
    kX9JihQE4X80jMzUKs7qnYcBIZLOPju7mLsn/u2GEnmjOGnUnOI2TrbvXkafHQbF4knZjBk/cOpafegoENH2
    Ys+CD7Ietaa40nlEk2kINcvjaPdxhQo0m5NAYPZoUullxyWN1Mi0xufRzFvHf+dX+GLQIHf/K7ggJHlZdf2MgSI
    qHgozcdosL9evEsPBSKJ9nJ+T34b4czsk2JApPHu+1Ir9n99U43sRoHgZB2XdO/1s9ge79LlZSlebyXrlT/hhCZJg8
    FIrMb0Bs0iRPYdIbIUBwfpZapY1oU8HomdbJtZMiDGam8I6PxUXI5TMRINL+n9qqbux7eD13jkTeKe56aVp+p
    AnSlu0K5WFQfGLG22m8bafi2IJIFLsRsXqhAWIqesc9OWN4q/VazXUV8dG6feEx0MRO219XFN5qaEqsPluB
    kRaVOzCtqEcSM/vtEKhcAR85xXbSjYnRrtVQhAnxmJrh2jUrmuPiMSCilwdSqe2ZOle0WG3yRnFvSdv8dRfzDH
    oFTRwoogh4r5Hli6Zt0wdDECC4exn9BNW7w7I5SOIV4UGX2e8dhNxD/NDUCQPGe6enxP8l+jSHqNAhNG9J6
    9zDR7VulY0QsHVzzFp1IbGmYwgREkCT4f35z4NxbpYQ1BE7/sWJL+gHfVIJHAosvPRXls7TPRL3fp6ESBEfRfJ
    ZWbn0Om8uScm107tdvIc5zDT9hr1lbkGAoSICG45pPHMz9jXXun14rbLtmQ1qTHYtMZrCBDpYu6uQKrfh+Ec
    mq28UdzO6/rmE+Rb1v3PD+shQAIuwUjX59QXIA2plfJGMVN7TUrZZFp1lT68AlqojX/elu9BeFnXDn8UCDFX
    UovutG2xTLrthgGRsdYc7FW3j6GVb6yebCZFaJDTanivJ+2T3ioyBkTcd4s0zc17/eyiWLq8UewmtPrNyS+bW6F17
    ggF14zcXEim7M0e2F6AAiGgqrLu7rhQVMi0VJoRWY2S6Kvqo+zuUZW+I1pqmGMr7li6zJTFVBIR64Qv6jaoGR
    u/B4Xg0yRFopXTZVjhRSi49o1NcTadguc5zstRIP8QlakhlipRyURU8n/L3zzIVd6RCcLvAAAAAEIFTkSuQmCC",
    "status": "A",
    "documentNumber": "8007",
    "signedInvoice":
    "ffrnTZAfrf2Jx1vNh9z3fbLyifKgz6iybyZ9g2j6c19FN6BN5/vv4kKBYbrw2TlbYrhNNH3/f58gVRKNfR92x/3JWJ
    IY2n1TDNOLivC3JT318+ZWPKE9d7OVxL2SirwDgY9bNWA81SZjGcV1hTBQbX8v39+dBVFJAV5XvRkvkSR
    sGgsz1YKanAgCuo5ID+X6BV++JQFavYYkgZKBxRyAkZJA+n2TPJlyeNCBx4SBi0Izvy9h/Tn0mvRUNaPvKAB
  }
]

```


xPqad/egMIHJdwtE0AX29oBFMngHMqBvyrh4eViSv6ZtP8phgv3OUJs5FMmXtIr/LfSdW5qTDcB+H8aVcfbIQ=
=",

"invoiceSignature": "xxxx",

},

{

"ruleError": [

{

"portion": "DocumentDetails",

"errorMessage": [

"DocumentDetails.date: Document creation time should be within 3 days"

]

},

{

"portion": "ReferenceDetails",

"errorMessage": [

"ReferenceDetails.id: Previous IRN in ReferenceDetail is not correct."

]

},

{

"portion": "SellerDetails",

"errorMessage": [

"SellerDetails.id: Seller's tax center is not correct"

]

},

{

"portion": "value details",

"errorMessage": [

"Exchange rate is not correct. "

]

}

],

"status": "ERROR",

"documentNumber": "101193"

},

{

"ruleError": [

{

"portion": "DocumentDetails",

"errorMessage": [

"DocumentDetails.date: Document creation time should be within 3 days"

]

},

{

"portion": "ReferenceDetails",

"errorMessage": [

"ReferenceDetails.id: Previous IRN in ReferenceDetail is not correct."

]

```
    },
    {
      "portion": "SellerDetails",
      "errorMessage": [
        "SellerDetails.id: Seller's tax center is not correct"
      ]
    },
    {
      "portion": "value details",
      "errorMessage": [
        "Exchange rate is not correct. "
      ]
    }
  ],
  "status": "ERROR",
  "documentNumber": "101194"
},
"conversationid" : "1234567899-1734682972171-de029d35-a527-4446-9cc1-36823a888c4c"
]
```

3. SECTION FIVE

3.2 INVOICE CANCELLATION

3.2.1 PURPOSE

The Electronics Invoice Management System (EIMS) and the Taxpayer System need to work together smoothly and efficiently to ensure that information is exchanged without any delays. To make this happen, it's essential for both systems to follow a set of predefined integration standards that ensure everything stays consistent and reliable.

This section is designed to give supplier and taxpayer clear and practical guidance on how to integrate with EIMS specifically for handling invoice cancellations. Invoice cancellations can happen for many reasons throughout the business process—whether it's due to changes in an order, customer disputes, or simple mistakes in the invoicing. EIMS is built to handle these situations seamlessly by providing a secure and easy-to-use API interface for cancellations.

By using this API, businesses can manage cancellations quickly and efficiently, ensuring compliance with regulations while keeping their records accurate and up to date. This section of the document aims to help developers implement this integration easily so businesses can streamline their processes and focus on what they do best.

The cancellation request is automatically processed if it adheres to the predefined daily and monthly cancellation limits and falls within the allowed time frame between invoice registration and cancellation. If the limits are exceeded or the request is outside this timeframe, it will undergo manual review.

3.2.2 OVERVIEW OF INVOICE CANCELLATION

The main purpose of the Electronics Invoice Management System (EIMS) is to handle essential tasks like registering, verifying, and canceling invoices. Among these, invoice cancellation happens quite often. This is because, in the day-to-day operations of a business, it's normal to encounter mistakes or the need for changes when generating invoices.

Invoice cancellations are initiated by Taxpayer (TP) systems for invoices that have already been registered in EIMS. To process a cancellation, the Taxpayer system needs to provide the Invoice Reference Number (IRN) of the invoice they want to cancel, along with a clear reason for why the cancellation is necessary.

The cancellation process follows a set of defined steps to ensure everything is handled accurately and in line with the rules. These steps help businesses make corrections or adjustments efficiently, without disrupting their workflows or falling out of compliance with regulations.

- 1 Request for invoice cancellation is initiated by Taxpayer system
- 2 EIMS validates the request JSON against cancellation schema
- 3 EIMS process the cancellation request processing it through various cancellation rules including but not limited to
 - Allowed number of cancellations in a day or a month
 - Allowed time between invoice registration and cancellation as specified by MoR
 - Allowed aggregate total value of invoices that can be canceled in a day or a month
 - If the rule validation of the request is successful, then the invoice cancellation confirmation response is sent to the Taxpayer system.
 - If the limits are exceeded or the request is outside this timeframe, it will undergo manual review

Taxpayer systems should fulfill the following when sending cancellation requests,

- Taxpayer systems must have been authenticated and a valid access token should be in the header
- Taxpayer systems must sign the request with their private key and include a valid certificate
- Taxpayer systems must encrypt the payload as specified in Security Compliance Document

3.2.3 HIGH-LEVEL ARCHITECTURE

3.2.3.1 System Components

- ✓ **Taxpayer System:** Generates invoice cancellation data, receives cancellation result
- ✓ **EIMS:** Applies the JSON schema rules, processes cancellation request

3.2.3.2 Data Flow

1. Taxpayer System generates cancellation request and submits to EIMS
2. EIMS validate the object against the schema.
3. EIMS processes the request and responds with cancellation result
4. Taxpayer System receives and stores the result

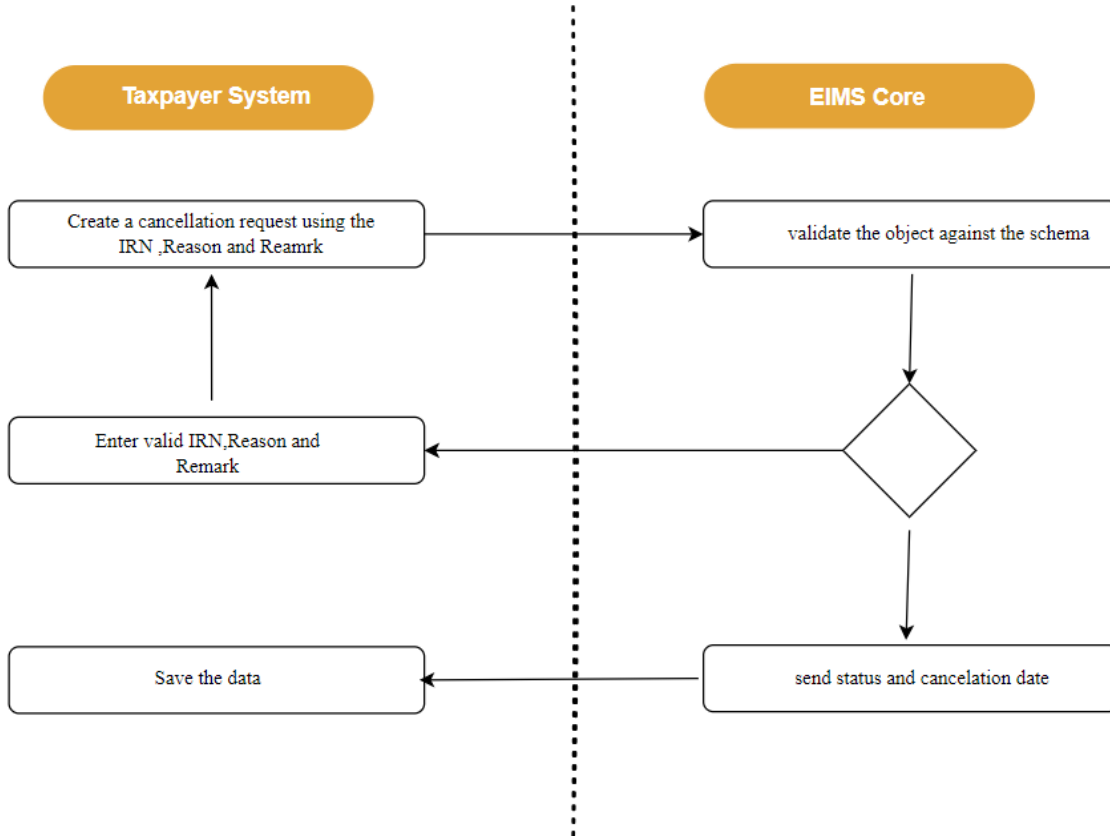


Figure 6: invoice Cancellation

3.2.4 SCHEMA OVERVIEW

To process an invoice cancellation, the request needs to follow a specific format outlined in the schema below. This schema ensures that the required information is provided in a clear and consistent way, allowing the cancellation to be handled smoothly and without any issues. Sticking to this format is important for maintaining accurate records and ensuring the systems communicate effectively.

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "MOR E-INVOICE",
  "$id": "http://www.mor.gov.et/einvoice",
  "description": "Invoice Cancellation schema",
  "type": "object",
  "properties": {
    "Irn": {
      "type": "string",
      "minLength": 64,
      "maxLength": 64,
      "description": "Invoice Reference Number"
    },
    "ReasonCode": {
      "type": "string",
      "minLength": 1,
      "maxLength": 1,
      "description": "Cancel Reason 1- Duplicate, 2 - Data entry mistake, 3- Order Cancelled, 4 - Others"
    },
    "Remark": {
      "type": "string",
      "minLength": 0,
      "maxLength": 100,

```

```

        "description": "Cancel Remarks"
      }
    },
    "required": [
      "Irn",
      "ReasonCode"
    ]
  }
}

```

As shown above the schema contains,

- Irn – Invoice registration number of the invoice to be canceled
- ReasonCode – a code for cancellation which can be one of the following

No	Reason Code	Description

- Remark – any description of the cancellation

Irn and ReasonCode are mandatory attributes.

Example of a cancellation request is show below,

```

{
  "Irn": "d0b10c49b4330d7cbc96f79116a6980f6e85ccea406e0b47196547c6b79334e6",
  "ReasonCode": "1",
  "Remark": "Repeated Error"
}

```

3.2.5 RESPONSES FOR CANCELLATION REQUEST

Just like any other system request, an invoice cancellation request can either go through successfully or fail for a variety of reasons. A successful request means everything was in order, and the system was able to process it without any issues. On the other hand, a request might fail due to problems like missing or incorrect information, formatting errors, or technical issues with the system.

The sections below break down what happens in both cases. They explain what a successful response looks like and how to handle error messages if something goes wrong. This way, you'll have a clear idea of what to expect and how to resolve any problems you might encounter.

3.2.5.1 Successful Cancellation Response

If the cancellation request is processed successfully, the response will follow the structure outlined in the schema below. This ensures that the information you receive is clear, organized, and easy to understand. The response will confirm that the cancellation went through and include any relevant details, so you'll have all the information needed to proceed or take further action if necessary. The structured response helps make sure everything is in order and that the process is transparent.

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "MOR E-INVOICE",
  "$id": "http://www.more.gov.et/einvoice",
  "description": "Invoice Cancellation Response schema",
  "type": "object",
  "properties": {
    "status": {
      "type": "string",
      "description": "Status of the cancellation request, mostly 'success' for successful requests."
    },
    "cancelationDate": {
      "type": "string",
      "format": "date-time",
      "description": "Cancellation date and time."
    }
  }
}
```



```

    }
  },
  "required": [
    "status",
    "cancellationDate"
  ]
}

```

Successful response example is given below,

```

{
  "status": "Success",
  "cancellationDate": "2024-12-22T15:45:17.769+00:00"
}

```

3.2.5.2 Failed Cancellation Responses

Cancellation responses may fail for various reasons, typically due to either a schema validation error or a cancellation rule validation error, as outlined in section 1.2. When the failure is caused by a schema validation error, the system will generate a response based on the following JSON schema. This schema specifies the exact structure and content of the error message, providing clear details on what went wrong, so that users or systems can address the issue and make the necessary corrections. By using this schema, the system ensures that the error is communicated effectively, allowing for a smooth resolution process.

```

{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "SchemaValidationErrorResponse",
  "type": "object",
  "properties": {
    "type": {
      "type": "string",
      "description": "The type of validation error (e.g., 'required')."
    },
    "code": {
      "type": "string",
      "description": "An error code representing the specific validation issue."
    }
  },
}

```

```
"message": {
  "type": "string",
  "description": "A descriptive error message regarding the validation failure."
},
"instanceLocation": {
  "type": "string",
  "description": "A JSON Pointer or path indicating where the error was found in the instance."
},
"property": {
  "type": "string",
  "description": "The specific property that caused the validation error."
},
"evaluationPath": {
  "type": "string",
  "description": "Path or pointer showing the evaluation context in the schema."
},
"schemaLocation": {
  "type": "string",
  "format": "uri",
  "description": "A URI reference to the schema rule that was violated."
},
"messageKey": {
  "type": "string",
  "description": "A key or code indicating the general type of validation error."
},
"arguments": {
  "type": "array",
  "description": "Additional argument details relevant to the validation error.",
  "items": {
    "type": "string"
  }
},
"required": [
  "type",
```

```

"code",
"message",
"instanceLocation",
"property",
"evaluationPath",
"schemaLocation",
"messageKey",
"arguments"
]
}

```

An example of schema validation error for invoice cancellation is show below,

```

{
  "type": "required",
  "code": "1028",
  "message": "$: required property 'Irn' not found",
  "instanceLocation": "$",
  "property": "Irn",
  "evaluationPath": "$.required",
  "schemaLocation": "http://www.mor.gov.et/einvoice#/required",
  "messageKey": "required",
  "arguments": [
    "Irn"
  ]
}.

```

In cases of rule validation error related to cancellation, the following schema will be used,

```

{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "MOR E-INVOICE ",
  "$id": "http://www.more.gov.et/einvoice",
  "description": "Invoice Cancellation Response schema",
  "type": "object",

  "properties": {
    "statusCode": {
      "type": "string",

```

```

        "description": "status of the cancellation request, mostly success in successful requests"
    },
    "message": {
        "type": "string",
        "description": "Rule violation error description"
    },
    "body": {
        "type": "array",
        "items": {
            "type": "object",
            "properties": {
                "code": {
                    "type": "string"
                },
                "property": {
                    "type": "string"
                },
                "message": {
                    "type": "string",
                }
            },
            "required": ["code", "property", "message"]
        },
        "required": [
            "status",
            "message"
        ]
    }
}

```

An example of rule violation error response is shown below,

```

{
  "message": "IRN not registered under the requesting tax payer's TIN ",
  "statusCode": "Error"
}

```

4 SECTION SIX

4.1 RECIEIPT

4.1.1 PURPOSE

The receipt module is a crucial component of the electronic invoice management system, playing a vital role in financial transactions. While an invoice serves as a request for payment, it does not guarantee that the payment has been completed. A receipt, on the other hand, acts as concrete proof that the payment has been successfully made, ensuring transparency, accuracy, and accountability in the transaction process.

This section offers detailed guidelines for software suppliers, taxpayers, and software developers on the proper procedures for submitting a receipt for an invoice that has already been registered within the electronic invoice management system. It outlines the necessary steps, compliance requirements, and technical specifications to ensure accurate submission and seamless integration within the system. By following these guidelines, all stakeholders can maintain proper financial records, enhance transaction transparency, and ensure adherence to regulatory standards.

4.1.2 OVERVIEW OF RECEIPT

Receipt management is a fundamental element of the electronic invoice management system, A receipt serves as concrete evidence that a payment has been successfully processed, offering guarantee to businesses, taxpayers, and regulatory bodies. This section will provide an in-depth exploration of key aspects of receipt management, including its importance, the step-by-step process for recording and submitting receipts for already registered invoice, compliance obligations, and best practices for seamless system integration. The following key points will be thoroughly examined to ensure a clear understanding and effective implementation.

4.2 HIGH-LEVEL ARCHITECTURE

4.2.1 SYSTEM COMPONENTS

- ✓ **Taxpayer System:** Generates receipt registration data, receives receipt registration result
- ✓ **EIMS:** Applies the JSON schema rules, processes receipt registration request

4.2.2 DATA FLOW

1. Taxpayer System generates receipt registration request and submits to EIMS.
2. EIMS validate the object against the schema.
3. EIMS processes the request and responds with receipt registration result.
4. Taxpayer System receives and stores the result.

4.3 SCHEMA OVERVIEW

To successfully register a receipt for an already registered invoice, it is essential to adhere to a predefined schema. This ensures a smooth and error-free registration process. Following the specified format outlined below is mandatory to guarantee the successful submission and validation of the receipt.

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "MOR E-INVOICE",
  "$id": "http://www.mor.gov.et/einvoice",
  "description": "Invoice Cancellation schema",
  "type": "object",
  "properties": {
    "ReceiptNumber": {
      "type": "string",
      "description": "Unique identifier for the receipt"
    },
    "ReceiptType": {
      "type": "string",
      "description": "Type of receipt, e.g., Sales Receipt"
    }
  }
}
```

```
"Reason": {
  "type": "string",
  "description": "Reason for issuing the receipt"
},
"ReceiptDate": {
  "type": "string",
  "format": "date-time",
  "description": "Date and time of the receipt issuance"
},
"ReceiptCounter": {
  "type": "string",
  "description": "Counter number for tracking receipts"
},
"ManualReceiptNumber": {
  "type": "string",
  "description": "Manually assigned receipt number, if applicable"
},
"SourceSystemType": {
  "type": "string",
  "description": "System type from which the receipt originates"
},
"SourceSystemNumber": {
  "type": "string",
  "description": "Unique identifier for the source system"
},
"ReceiptCurrency": {
  "type": "string",
  "description": "Currency code of the receipt (e.g., USD, EUR)"
},
"ExchangeRate": {
  "type": ["number", "null"],
  "description": "Exchange rate applicable at the time of transaction"
}
```

```
},
"CollectedAmount": {
  "type": "number",
  "description": "Total amount collected in the transaction"
},
"SellerTIN": {
  "type": "string",
  "description": "Taxpayer Identification Number of the seller"
},
"Invoices": {
  "type": "array",
  "description": "List of invoices associated with this receipt",
  "items": {
    "type": "object",
    "properties": {
      "InvoiceIRN": {
        "type": "string",
        "minLength": 64,
        "maxLength": 64,
        "description": "Invoice Reference Number (IRN)"
      },
      "PaymentCoverage": {
        "type": "string",
        "description": "Specifies if the invoice payment is full or partial"
      },
      "InvoicePaidAmount": {
        "type": "number",
        "description": "Amount paid for the invoice"
      },
      "DiscountAmount": {
        "type": ["number", "null"],
        "description": "Discount amount applied, if any"
      }
    }
  }
}
```



```
    },
    "RemainingAmount": {
      "type": ["number", "null"],
      "description": "Remaining amount to be paid, if applicable"
    },
    "TotalAmount": {
      "type": "number",
      "minimum": 0,
      "description": "Total amount of the invoice"
    }
  },
  "required": ["InvoiceIRN", "PaymentCoverage", "InvoicePaidAmount", "TotalAmount"]
},
"TransactionDetails": {
  "type": "object",
  "description": "Details of the transaction",
  "properties": {
    "ModeOfPayment": {
      "type": "string",
      "description": "Payment method used (e.g., CASH, CARD)"
    },
    "ChequeNumber": {
      "type": ["string", "null"],
      "description": "Cheque number if payment was made via cheque"
    },
    "CPONumber": {
      "type": ["string", "null"],
      "description": "Customer Purchase Order number, if applicable"
    }
  },
  "DocumentNumber": {
    "type": ["string", "null"],
```

```
    "description": "Reference document number, if applicable"
  },
  "CollectorName": {
    "type": "string",
    "description": "Name of the person collecting the payment"
  },
  "PaymentServiceProvider": {
    "type": "string",
    "description": "Payment service provider facilitating the transaction"
  },
  "OtherPaymentServiceProviderName": {
    "type": ["string", "null"],
    "description": "Additional payment service provider name, if applicable"
  },
  "AccountNumber": {
    "type": "string",
    "description": "Account number used for the transaction"
  },
  "TransactionNumber": {
    "type": "string",
    "description": "Unique identifier for the transaction"
  }
},
"required": ["ModeOfPayment", "CollectorName", "PaymentServiceProvider", "AccountNumber",
"TransactionNumber"]
},
"required": [
  "ReceiptNumber", "ReceiptType", "Reason", "ReceiptDate", "ReceiptCounter",
  "SourceSystemType", "SourceSystemNumber", "ReceiptCurrency", "CollectedAmount",
  "SellerTIN", "Invoices", "TransactionDetails"
]
```

```
}
```

Example of a receipt registration request is show below,

```
{
  "ReceiptNumber": "REC1234567890123466",
  "ReceiptType": "Sales Receipts",
  "Reason": "Payment for goods purchased",
  "ReceiptDate": "2025-01-10T14:30:00.000+03:00",
  "ReceiptCounter": "12345",
  "ManualReceiptNumber": "98765",
  "SourceSystemType": "MAX",
  "SourceSystemNumber": "112345678",
  "ReceiptCurrency": "USD",
  "ExchangeRate": null,
  "CollectedAmount": 240.0,
  "SellerTIN": "1234567899",
  "Invoices": [
    {
      "InvoiceIRN":
"4c484b18fb0770c9dc3290a4dd15999f8a682a7d79e5610d1d89bd79007c0d7r",
      "PaymentCoverage": "FULL",
      "InvoicePaidAmount": 1000.00,
      "DiscountAmount": null,
      "RemainingAmount": null,
      "TotalAmount": 240.0
    }
  ],
  "TransactionDetails": {
    "ModeOfPayment": "CASH",
    "ChequeNumber": null,
    "CPONumber": null,
    "DocumentNumber": null,
    "CollectorName": "Jane Doe",
    "PaymentServiceProvider": "Bank",
    "OtherPaymentServiceProviderName": null,
    "AccountNumber": "123456789",
    "TransactionNumber": "TRX987654321"
  }
}
```

4.3.1 RESPONSES FOR RECEIPT REGISTRATION

The receipt request can generate two types of responses: a successful response or a failure response. A successful response is issued when all mandatory fields are provided, and the request adheres to the predefined schema and validation rules. This ensures that the receipt is properly registered without any issues. On the other hand, a failure response occurs if there is an error in the request, such as missing required fields, incorrect data formatting, or a violation of the

established validation rules. In such cases, the request is rejected, and appropriate error messages are returned to indicate the specific issues that need to be corrected before resubmission.

The following sections provide a detailed breakdown of both scenarios. They outline the structure of a successful response and offer guidance on handling error messages in case of issues. This ensures you have a clear understanding of what to expect and how to troubleshoot any potential problems effectively.

4.3.1.1 Successful Receipt Registration Response

If all required attributes are provided and the predefined validation rules are successfully met, the system will generate the following expected response. This response serves as confirmation that the receipt request has been processed correctly and complies with all necessary requirements. It ensures that the transaction is valid and recorded without errors, allowing for smooth and efficient processing.

```
{
  "statusCode": 200,
  "message": "Success",
  "body": {
    "status": "A",
    "rrn": "8533e509754800709b8de248297985da1cbc8eaeb734add884d87875e5577589",
    "qr":
      "iVBORw0KGgoAAAANSUgAAAlgAAAIAQAAAAACWHaVxAAAIEIEQVR4Xu2aQW7cSAxF2eiFlz6CjqKJS
      UfTUXQEL71odA3fJ6vjODNAhMlihiAXkVUiX2XxQf4q28afC/u+8C+iWdeiWdeiWdeiWdeiWdeiWdei
      WdeiWdeiWdfiv836MOJtnOt43j/f/eFvH8uxPc3ex9h9kYTxeFP6w0jxN09R5dKseixe/JM55C7yk5R13NjHH4
      9AOvk0SrJgJGObb82qxjrspk/mLGTGw6tjA9dXLJqkBPlcd6Ucm5TYrKosX9tBnhQp3/WIN1imDnR+0ZeLrInF
      WZTpfczWQ6L09ek/brwv3mVCX7fxyVuzqrJGziEyXENRNPfbzqGYSilv+JZ/mEPNKsGiFJ0cm1rPhYcqM/Xbj/
      8N6xWHuQ1BLs/oJLkBi8gMKQGJngPra3GzKrHQL2d4bNFQDJakg/+waVEIN6rGxIoN1IHem1WPlcLkZMnzS
      gO1QcQ+V6dXUajShNLuzarKotENEQi5HlcmUaV6mg9pkY0dPQ9+TZbT7NqseKTSymrY+RwC+qJWlw4vLxF
      Lmrzb4a+BplBSFKzKrDkPyw+KYMDilqPxxmTx/exmcJpR4VDZ5iMZimxCltmI1OiFyBhRnUzDuSu7fRV24Vf5
      dtQk1I0qxJLTYNqDaDUFzfjKpOX1RzScGLyUCRTcmzap1n1WNFCMB5iSTX+g9rL8tJXth4U5RtoHFH3TV/NKs
      GKX5ZJX0P+I04tvqgHLFLQL1gKDK7rCxO7NKsgK22IHswhoywMi8FCbdPE8vDFEJbUFpBmVWLRbNCXZ4Rc9
      iRn6wkbEqLzVwaQvzGqfNeXp2IWJRahTnLoysMfDCBX1BNhuZTiYKMdncwA0kMu5nvPaVYdgaQJs9QJ3HI
      CA1JUS6InEMxqhckQdsS7OqsXjJhoKwUE0kSmaaQ1JUQAi9aVThWyKaVYnIXSYTd9nWGDlZUeTQOCcdKEf
      VExNrpvNNswqycCOvaoQViyo7SEJ0dJnBqNIFiEUH0q7NqsgafCMf+3nYU6eWaUM0ctZMEdmMZoPPDV02
      qxpLazxW3XuyFI7D17nrmF42ZIY3OaOAlG/6aLYNVs4hpguJrKvZplboMrbl0rJrJCogEuScUc0qxfqqrztdhuMK
      1cp3L8vbMRWVEysscufhZhVkoUliUBTtBYgCte2aQ5IBWwpFPXJFuzYrFsqwEJZOMg+0phuvealZHFdiA1laH
```

PkeNaHCgtWaVZRnxJn0Zp5Y95pDKTmOD4Q8yF1lauZhcpPU0qxrz0SzkWzrkWzrkWzrkWzrkWzrkWzrkWzrkWzrkWzrkWzrsVfbY328sxI5vYAAAAASUVORK5CYII=

}

If the request to register the receipt fails due to not meeting the predefined required attributes and the established rules, the following responses are expected. These responses will provide details about the specific reasons for failure and highlight which required criteria or rules were not fulfilled during the process.

```
{
  "statusCode": 406,
  "message": "NOT_ACCEPTABLE",
  "body": {
    "status": "F",
    "unknownInrn": [
      "4c484b18fb0770c9dc3290a4dd15999f8a682a7d79e5610d1d89bd79007c0d7r"
    ],
    "dataMissing": [],
    "ruleErrorDto": [
      {
        "portion": "Receipt",
        "errorMessage": [
          "Currency should be ETB, USD, CAD",
          "Source System Type should be either POS, MAN, CRM, EFD, SYS",
          "Invoice currency is not found: Invoice currency is not found"
        ]
      },
      {
        "portion": "TransactionDetails",
        "errorMessage": [
          "Payment mode should be one of CASH, ADVANCE, or CREDIT"
        ]
      }
    ]
  }
}
```

5 SECTION SEVEN

5.1 SECURITY

5.1.1 PURPOSE

One of the biggest concerns when integrating the Electronic Invoice Management System (EIMS) with taxpayer systems is security. Protecting sensitive financial data and ensuring the integrity of the systems is critical for smooth and reliable operations. That's why security is a top priority in making sure the EIMS and taxpayer systems work together seamlessly.

This section is designed to provide clear guidance on the key security aspects, including authentication, authorization, and the use of digital certificates. These processes are essential to ensure secure communication and protect the systems involved. By following the guidelines outlined here, businesses can confidently integrate with EIMS while maintaining a safe and compliant environment for managing electronic invoices.

5.1.2 OVERVIEW

This section outlines the following essential requirements for integrating with the Electronic Invoice Management System (EIMS):

- **Authentication Requirements:** Detailed guidelines on the authentication processes and protocols necessary for secure integration with EIMS.
- **Payload Encryption Requirements:** Specifications on how data payloads should be encrypted to ensure the confidentiality and integrity of information exchanged with EIMS.
- **Digital Certificate Requirements:** Standards and guidelines for using digital certificates to establish trusted and secure communication with the system.

Additionally, the document provides step-by-step instructions on how to obtain the necessary credentials, encryption keys, and digital certificates. Each section includes comprehensive details to help users meet these requirements and successfully integrate their systems with EIMS while maintaining robust security.

5.1.3 AUTHENTICATION IMPLEMENTATION

The Electronic Invoice Management System (EIMS) is built to allow taxpayer systems to connect directly using APIs for tasks like registering, canceling, and validating invoices. However, before any of these actions can be performed, the taxpayer system needs to authenticate with EIMS. This is done by providing specific credentials, including a client ID, client secret, API key, and the Taxpayer Identification Number (TIN).

Taxpayers with a valid TIN can easily get these credentials through the taxpayer email or phone number. It's important to create unique credentials for every system that will connect directly to EIMS. Once generated, these credentials need to be set up within the systems intended to handle tasks like invoice registration, cancellation, or verification. Following this process helps ensure the systems are secure, efficient, and ready to integrate smoothly with EIMS.

5.1.3.1 Authentication Request and Response

5.1.3.2 Request

Every request sent to the Electronic Invoice Management System (EIMS) must include a digital signature. This helps ensure that the data being transmitted is secure, authentic, and hasn't been tampered with. (You can find detailed guidance on how to obtain and use a digital signature in a separate section.)

The schema shown below gives an overview of how these requests should be structured. Following this framework is crucial for ensuring smooth communication and successful integration with EIMS.

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "type": "object",
  "properties": {
    "request": {
      "type": "object",
      "properties": {
        "clientid": {
          "type": "string"
        },
      },
      "clientsecret": {
```



```

        "type": "string"
    },
    "apikey": {
        "type": "string"
    },
    "tin": {
        "type": "string"
    }
},
"required": [
    "clientid",
    "clientsecret",
    "apikey",
    "tin"
]
},
"signature": {
    "type": "string"
},
"certificate": {
    "type": "string"
}
},
"required": [
    "request",
    "signature",
    "certificate"
]
}

```

The request property has the following entries,

- clientid – a unique client id that is provided to the Taxpayer system in Taxpayer portal
- clientsecret – a client secret provided to the Taxpayer system on Taxpayer portal
- apikey – a unique API key that is provided to the Taxpayer that should be used in all Taxpayer systems
- TIN – Taxpayer Identification Number

Communication Type	REST, Post
URL	x.x.x/authenticate
JSON Schema	authenticaitonSchema
Version	1

5.1.3.3 Response

When you make an authentication request, it can either be successful or return an error response, depending on whether the provided credentials meet the system's requirements. This section starts by explaining what a successful authentication response looks like, followed by a breakdown of possible error responses and what they mean.

The schema below shows the structure of a successful authentication response, highlighting the key elements included. Understanding this schema will help you interpret the system's feedback and ensure a smooth authentication process.

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "type": "object",
  "properties": {
    "data": {
      "type": "object",
      "properties": {
        "accessToken": {
          "type": "string"
        },
        "refreshToken": {
          "type": "string"
        },
        "encryptionKey": {
          "type": "string"
        },
        "expiresIn": {
          "type": "integer"
        }
      }
    },
    "required": ["accessToken", "refreshToken", "encryptionKey", "expiresIn"]
  }
}
```

```
},  
  "status": {  
    "type": "string"  
  }  
},  
  "required": ["data", "status"]  
}
```

Contained in the response are

- accessToken – this is a bearer token that has to be part of the header subsequent API calls which will be used for successful authentication
- refreshToken – a token that can be used to get a new accessToken before it expires so that Taxpayer systems do not have to re-login when accessToken nears expiry time
- encryptionKey – payload encryption key that will be used for communication between the Taxpayer system and EIMS
- expiresIn – the number of seconds before the accessToken expires
- status – status of the request, success or failure, for successful authentication this should always be 200

For requests that are not successful, error reply of the following schema will be responded,

```
{  
  "$schema": "http://json-schema.org/draft-07/schema#",  
  "type": "object",  
  "properties": {  
    "message": {  
      "type": "string"  
    },  
    "code": {  
      "type": "integer"  
    },  
    "details": {  
      "type": "array",  
      "items": {  
        "type": "object",  
        "properties": {  
          "field": {  
            "type": "string"  
          },  
          "errorMessage": {  
            "type": "string"  
          }  
        },  
        "required": ["field", "errorMessage"]  
      }  
    }  
  }  
}
```

```
},  
  
"status": {  
  
"type": "string"  
  
}  
  
},  
  
"required": ["message", "statusCode", "details", "code"]  
  
}
```

The contents of the error response are,

- message – a description of the error in a text format
- code – a numerical code of the error that is generated by EIMS which Taxpayer systems can use for better error detection currently defined are 4001 when required values are missing and 4110 when authentication credentials are invalid
- details – an array of error lists which give detailed error information
- status – error response status with http error code, for unsuccessful authentication this is usually 40X, where X varies depending on the error

5.1.4 Token Refresh

To avoid disruptions, taxpayer systems can send token refresh requests before the current access token expires. This helps maintain a smooth and uninterrupted connection with the system.

For a token refresh request to be processed successfully, it needs to follow the structure outlined in the schema below. The schema specifies the required components and format, ensuring the request aligns with the system's standards and works as intended.

```
{  
  "$schema": "http://json-schema.org/draft-07/schema#",  
  "type": "object",  
  "properties": {  
    "refreshToken": {  
      "type": "string"  
    }  
  },  
  "required": ["refreshToken"]  
}
```

The token refresh request only requires one parameter: the refreshToken, which is provided when the authentication request is successful. This token allows you to get a new access token without having to log in again.

If the token refresh is successful, the system will send a JSON response that follows the same format as the successful authentication response. However, if the refreshToken is invalid or something goes wrong, the system will return an error message in a format similar to the authentication error responses, making it easy to understand what went wrong.

5.1.5 Payload Encryption

To ensure greater security, all communications between taxpayer (TP) systems and the Electronic Invoice Management System (EIMS) will have their payloads encrypted, except for authentication and token refresh requests. This extra layer of protection will help keep the data secure during most interactions between the two systems.

As a result, the way messages are exchanged between the systems will follow the structure outlined below, making sure that the information stays private and safe while being transmitted.

```
{  
  
    'data' : "cipher text"  
  
}
```

The cipher is created using a symmetric encryption algorithm of -----, with the encryption key provided during successful authentication. This key is valid only for the current session and will be refreshed each time a new authentication takes place or when the access token is renewed. This approach ensures that the encryption key is regularly updated, helping to maintain strong security throughout each session.

5.1.6 DIGITAL CERTIFICATE REQUIREMENTS OF EIMS

The Electronic Invoice Management System (EIMS) requires that all taxpayer (TP) systems connected to it digitally sign every request they send. In addition, Taxpayer systems must verify the authenticity of the responses from EIMS by checking its digital signature. To do this, Taxpayer systems will need to obtain a digital certificate from the Ministry of Revenue (MoR), through a process that will be defined at a later stage. This certificate is crucial for ensuring secure communication and maintaining the trustworthiness of the data exchanged between the systems.

The digital certificate generation requires the following steps,

1. Taxpayer system generates a private and public key pair. The keys generated should meet the requirements of FIPS 186 (latest version) and the public key will be validated against ECC Full Public Key validation.
2. Taxpayer systems should generate a certificate signing request with the following contents,
3. Taxpayers provide the Certificate Signing Request (CRS) to
4. Taxpayer download and use digital certificates as per the requirement

CRS Input	X509 Certificate Field	Description	Accepted Input
countryName	x509.subject.country	Ethiopia	ET
stateOrProvinceName	x509.subject.state	Region name	As per the master code declared by MoR
organizationName	x509.subject.organization	Legal name of the organization	Free text input
SerialNumber	x509.subject.serial		
commonName	x509.subject.name	TIN number of tax payer	The Taxpayer Identification Number

Taxpayer systems will need to use the public key from the provided digital certificate to sign the JSON message contents whenever they send requests to EIMS. This helps ensure that each message is securely authenticated and verifies its integrity. By including the digital signature, it guarantees that the message hasn't been tampered with during transmission and confirms that it comes from the authorized taxpayer system. This process is vital for ensuring secure, trustworthy communication between the Taxpayer systems and EIMS.

1. shall comply with the requirements of FIPS 186-4 (Digital Signature Standard)
2. approved algorithms shall include ECDSA over the NIST P-256 curve and RSA with a 2048-bit key

6 ANNEX 1

የገንዘብ መቀበያ ደረሰኝ (Receipt Invoice)

Type

IRN: የደረሰኝ ምዝገባ ቁጥርMRC: ሲብስታን ቁጥር

ከ
From የግብር ከፋይ መታወቂያ ቁጥር ላይ የተጠቀሰው ስም

ለ
To _____

አድራሻ: ከተማ
Address: City/Town
ዞን / ክ/ከተማ
Zone/Sub city
ወረዳ
Woreda
ቀበሌ
Kebele
የቤ/ቁ
H/No

አድራሻ : ከተማ
Address: City/Town
ዞን / ክ/ከተማ
Zone/Sub city
ወረዳ
Woreda
ቀበሌ
Kebele
የቤ/ቁ
H/No

የሻጭ ተ.እ.ታ ቁጥር
Seller's VAT Reg.No
የሻጭ የግብር ከፋይ መለያ ቁጥር
Seller' TIN
ለተ.እ.ታ የተመዘገበበት ቀን
Date of VAT Registration

የገዥ ተ.እ.ታ ቁጥር
Customer's VAT Reg.No
የገዥ የግብር ከፋይ መለያ ቁጥር
Customer's TIN
ለተ.እ.ታ የተመዘገበበት ቀን
Date of VAT Registration

የክፍያ ምክንያት

Payment Description: _____

ተ/ቁ S.N	የደረሰኝ ቁጥር Invoice Doc. Number	የደረሰኝ ጠቅላላ መጠን Total Invoice Amount	የገንዘብ ዓይነት Invoice Currency	ተይዞ የቀረ መጠን Withheld Amount	የተሰጠ ቅናሽ Discount Given	የተከፈለ መጠን Collected Amount
B.						
C.						
D.						
E.						
F.						
Total Invoice Amount {{ Currency}}/ጠቅላላ የደረሰኝ መጠን {{የገንዘብ}}						#,###.##
Total Discount Given / ጠቅላላ የተሰጠ ቅናሽ						#,###.##
Total Withheld Amount / ጠቅላላ የተያዘ መጠን						#,###.##
Collected amount {{Currency}} / የተሰበሰበ መጠን {{የገንዘብ አይነት}}						#,###.##
Exchange Rate to Birr						

ጠቅላላ ዋጋ በፊደል ብር

In Words **Birr** : _____

የክፍያ ሁኔታ Mode of Payment:	Type/Method አይነት _____	Reference ማጣቀሻ _____
	አገልግሎት ሰጪ Service Provider _____	Ticket ቲኬት _____
		From ከ _____
		To እስከ _____

ያዘጋጀው ስምና ፊርማ Prepared By _____	የተቀበይ (ሻጭ) ስምና ፊርማ Receiver Name & Signature _____ {{Cashier}}
በ{{ታክስ ባለስልጣን}} በ {{ታክስ ሴንተር}} ቅ/ፅ/ቤት በ {{ቀን}} ቀን {{ወር}} ወር {{ዓ/ም}} ዓ/ም የኮምፒውተር ፕሮግራም ጥቅም ላይ እንዲውል የተፈቀደ።	

DRAFT